Combustion and Flame 206 (2019) 1–20

Contents lists available at ScienceDirect

Combustion and Flame

journal homepage: www.elsevier.com/locate/combustflame

A chemical kinetic modeling study of indene pyrolysis

Hanfeng Jin[a], Lili Xing[b], Junyu Hao[a], Jiuzhong Yang[c], Yan Zhang[d], ChuangChuang Cao[d], Yang Pan[c], Aamir Farooq[a,*]

[a] Clean Combustion Research Centre, King Abdullah University of Science and Technology, Thuwal 23955-6900, Saudi Arabia
[b] Energy and Power Engineering Institute, Henan University of Science and Technology, Luoyang, Henan 471003, China
[c] National Synchrotron Radiation Laboratory, University of Science and Technology of China, Hefei, Anhui 230029, China
[d] Key Laboratory for Power Machinery and Engineering of MOE, Shanghai Jiao Tong University, Shanghai 200240, China

ARTICLE INFO

Article history:
Received 18 January 2019
Revised 18 April 2019
Accepted 18 April 2019

Keywords:
Indene mechanism
Flow reactor pyrolysis
SVUV-PIMS
Polycyclic aromatic hydrocarbons
Resonantly stabilized radical

ABSTRACT

An improved polycyclic aromatic hydrocarbon (PAH) model is developed to predict the decomposition of indene and the formation of large PAHs under pyrolytic conditions. This model is developed based on experimental study of pyrolytic kinetics of indene in a flow reactor at low and atmospheric pressures (30 and 760 Torr) by using synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS). A general map of PAH growth is presented according to the observations in this study and those in literature. Indene dissociates via indanyl forming mono-cyclic aromatics and small intermediates, while its dominant decomposition product is indenyl. As a resonantly stabilized radical, indenyl serves as a platform molecule in PAH growth process which links small unsaturated hydrocarbons and mono-aromatic species to multi-cyclic ones. Reactions of indenyl radical are proposed to form commonly studied and recently observed PAHs. Rate constants of these reactions are evaluated by analyzing literature data of rate constant measurements, quantum chemical calculations and analogy to cyclopentadienyl radical. The main PAH formation pathways are the bi-molecular addition reactions of indenyl radical with indene and a series of intermediates, forming C_{10}–C_{18} and larger PAHs. Meanwhile, radical chain reactions provide huge passage for PAH growth form one resonantly stabilized radical (RSR) to larger ones. Particular contribution has been found from the reactions of RSRs that have five-member ring in their molecular structures, such as fluorenyl, benz-indenyl, cyclopenta-phenanthrenyl and benzo-fluorenyl.

1. Introduction

Polycyclic aromatic hydrocarbons (PAHs) and soot are formed as a result of incomplete combustion, and contribute to particulate matter (PM) in our atmosphere. Human exposure to high density of PM enhances the chances of heart and lung diseases [1,2]. The investigation of the formation of PAHs and soot has, therefore, gained increasing attention since 1990s [3–6]. Various PAH models have been developed with a large array of possible reactions [7–15]. Herein, we briefly review the reaction kinetics for bicyclic aromatic species in these models.

In a general view of the history of PAH model development, modelers generally aim to fill the blanks in the giant map of PAH species. Wang and Frenklach [16] proposed hydrogen-abstraction-carbon-addition (HACA) mechanism in 1994, which was later supported by a number of experimental and theoretical investigations [13,17–20]. Models developed by Wang and Frenklach [14] and Appel et al. [13] were then reported to describe the formation and growth of PAHs in flames fueled by C_2 hydrocarbons, such as ethane, ethylene and acetylene. These models focused on HACA routes, and had relatively less consideration on toluene, indene, methylnaphthalene, etc., which may have high quantities in flames [15,21–23]. Later, Marinov et al. [12] extended the PAH model to cyclopenta[*cd*]pyrene, and Richter et al. [11] described the PAH growth up to coronene. They even proposed reaction pathways to form C_{60}–C_{70} fullerene. These two works represented early investigations that not only included HACA mechanism but also the contributions from radical combination routes [12,24–29]. Propargyl and cyclopentadienyl radicals were then considered to be efficient contributors to the formation of first and second aromatic ring, respectively [4,15,30–38]. Slavinskaya et al. [10,39] reviewed previous investigations and developed a PAH model up to five aromatic rings. Later, Shukla and Koshi [40] added phenyl and methyl addition reactions in their model to extended the HACA mechanism. Hansen et al. [41] proposed repetitive addition of methyl and

* Corresponding author.
E-mail addresses: xingll2018@haust.edu.cn (L. Xing), aamir.farooq@kaust.edu.sa (A. Farooq).

https://doi.org/10.1016/j.combustflame.2019.04.040
0010-2180/

acetylene for PAH growth. A series of radical-involved PAH formation pathways have been proposed and calculated [42–54]. Finally, a number of PAH models [7–9,19,30,55–58] have thus been developed and validated. We will not detail those studies here but will instead discuss some recent works, particularly those closely related on indene.

A recent study of Johansson et al. [59] has confirmed the critical role of resonantly stabilized radicals (RSR) in the growth of PAH and soot. Some recent theoretical calculations also revealed details of the reaction mechanisms of RSRs combination [47,48]. All these kinetic developments provide a good opportunity to model the PAH growth process from bicyclic aromatics to very large ones. Indene (C_9H_8) is a typical bicyclic aromatic species, which is an important transmitting platform in the aromatic growth sequence. It could produce indenyl (C_9H_7) radical *via* H-abstraction or H-elimination reactions. Indenyl radical has a resonantly stabilized molecular structure, and is, therefore, very participative in radical-radical and radical-neutral recombination reactions [59,60]. Kinetic modeling study on indene can help reveal not only indenyl-involved PAH growth pathways but also the aggregation of small molecules formed by the decomposition of indene.

There are only a few literature studies related to indene chemical kinetics [61–67]. Badger and Kimber [67] performed gas chromatography (GC) measurements of indene pyrolysis, and observed significant amount of benzofluorenes, benz[*a*]anthracene, benzophenanthrene and chrysene. Laskin and Lifshitz [62] measured thermal decomposition products of indene behind reflected shock waves also by GC. Indenyl and indanyl radicals were recommended in their work as the main pathways of indene consumption. Benzene, toluene and phenylacetylene were proposed to be formed *via* five-member ring dissociation of indanyl radical, while the decomposition of indenyl formed 1-ethynyl-cyclopenta-2,4-dienide (C_7H_5) and cyclopentadienyl (C_5H_5) radicals. Flash pyrolysis of 2-ethynyl-toluene ($C_2HC_6H_4CH_3$) was studied in the work of Ajaz et al. [61], and that of 3-methyl-4-(2′- methyl)benzylideneisoxazol-5(4H)-one by Wentrup et al. [63]. The quantum chemical calculations of Wentrup et al. [63] explained the interconversion between indene and 2-ethynyl-toluene by Roger-Brown rearrangement. Lifshitz et al. [66] investigated the pyrolysis of 2-methyl-indene in a shock tube, and described the mechanism of indenyl radical according to their experimental observations. Literature quantum chemical investigations of indene have focused on the addition reactions of indenyl. The addition of acetylene on indenyl radical was calculated in the work of Cavallotti et al. [64]. In a recent study, Mebel et al. [53] proposed the conversion from indene to naphthalene, emphasizing that 1-methyl-indene and 1-methyl-indenyl are critical intermediates in this process. The reaction of indenyl radical was investigated by Lu and Mulholland [65] *via* quantum chemical calculations and species identification experiment. They proposed the product of this 9-step mechanism to be chrysene, similar to the formation of naphthalene from cyclopentadienyl. Calculations of Wentrup et al. [63] also showed that the combination of indenyl could produce bi-indene and chrysene.

In the current work, detailed experimental data of indene pyrolysis in a flow reactor are reported and used to propose a new model for indene kinetics. Experiments were performed with synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) combined with molecular-beam sampling technique [68,69]. Indene-contributed PAH sub-mechanism is constructed based on the present experimental measurements and quantum chemical calculations. The major improvement in the present model is the inclusion of semi-detailed PAH growth pathways *via* indenyl-involved reactions. The model is validated against experimental data from this study and literature. Important intermediates and reactions are identified for their critical role in the decomposition and ring expansion of indene.

2. Experimental details

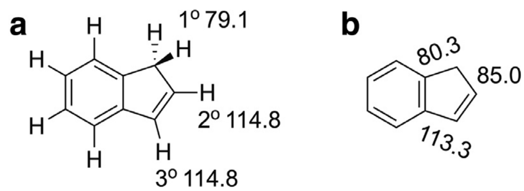
Indene pyrolysis was studied in a flow reactor at an undulator-based VUV beamline (BL03U) of Hefei Light Source (HLS II), China. This new pyrolysis facility and the experimental evaluation process have been introduced in detail in previous works [68,69]; only a brief description will be given here. The flow reactor is composed of a home-made furnace and an α-type alumina ceramic tube (inner diameter 7.0 mm, wall thickness 1.5 mm). Electrically heated wire is uniformly twined in the inner layer of the furnace, and the outermost layer acts as a heat shield. The total length of the heating zone is ~220 mm. A tungsten–rhenium (W–Re) thermocouple is positioned at the middle region to control the reactor temperature. For the experiments reported here, mixtures of argon and indene were fed into the alumina tube in the gas phase, and the pyrolysis species were sampled 10 mm downstream from the outlet of the ceramic tube by a quartz nozzle. The forming molecular beam was passed through a skimmer, and ionized by an orthogonal tunable VUV light. Ions were detected by a time-of-flight mass spectrometer (TOF-MS) *via* an ion introducer.

Since temperature along the centerline of the flow reactor is a key parameter for the pyrolysis process in the flow reactor, accurate temperature profiles were measured by an S-type thermocouple before/after each measurement. Details about the temperature measurement are provided in *SMM2* Section 1.1. The measured temperature profiles are given in *SMM1*, which are used in the simulation of the flow reactor [70].

Liquid indene (purity 97%) is injected in a vaporizer at 485 K before being fed into the flow reactor. It is regulated by a high-pressure chromatography pump, while argon (purity 99.99%) is regulated by an MKS mass flow controller. The molar composition of the mixture is 0.32% indene / 99.68% argon, and the total flow rate of the mixture is 1.5 SLM. Mass spectra were obtained at two pressures, 30 and 760 Torr, and temperature was varied in steps of 25 K over 975–1450 K. The residence time is estimated to be 20 and 500 ms at 30 and 760 torr, respectively. The uncertainty of SVUV-PIMS measurements comes primarily from the sampling process and the photoionization cross-sections (PICS) of the measured species [68]. Generally, the measurement uncertainty is ±10% for species with direct PICS measurement, ±25% for stable species with PICS available in literature or database, and a factor of 2 for free radicals (usually underestimated due to annihilation in the sampling process) and species with estimated PICS. Further details of the experimental methodology, such as purity of indene, mole fraction evaluation, carbon balance at each temperature, and PICSs of various species, are given in *SMM2*.

3. Theoretical calculations

Quantum chemical calculations were employed in this work for the molecular structures, bond energies and reaction pathways. Geometry optimizations, electronic energies, and frequencies were computed using M08-HX method [71] with the MG3S basis set [72]. The MG3S basis set is identical to the 6-311+G(2df,2p) basis set for H, C, and O atoms [73]. Self-consistent field calculations and geometry optimization were performed with tight convergence criteria. Density functional integrations were carried out by a grid of 99 radial shells per atom with 974 angular points per shell. All calculations were performed using the Gaussian 16 software [74] with a self-developed MN-GFM6.7 module [75]. M08-HX/MG3S method was selected as an affordable electronic structure method because it has been shown to have uncertainty below 1 kcal/mol [76]. Cartesian coordinates (in Å) and absolute energies of M08-HX/MG3S optimized geometries are collected in *SMM3*.



Scheme 1. (a) C–H and (b) C–C bond energies of indene, calculated at 298 K. Unit: kcal/mol.

4. Improved kinetic model

We roughly divided indene model into two parts: the decomposition routes and the aromatic growth routes. The first part of the current model is referred to the base models from Yuan et al. [9,56], Pousse et al. [77] and Lifshitz et al. [66]. The second part of the model is the PAH growth from indenyl radical. The model of Yuan et al. is adopted as the base model, and then we built up reaction blocks for large PAHs. We intended to keep the rate constants in the base model, and added necessary PAH formation reactions driven by indenyl. Rate constants of these reactions are taken either from values in the literature [40,47,50,62,65,66,78] or analogized to similar reactions of cyclopentadienyl radical [46,48,51,64,79].

4.1. Indene (C_9H_8) decomposition reactions

4.1.1. Indenyl (C_9H_7) formation

Indenyl radical is the most critical intermediate in the growth of PAH from indene. Indenyl may be formed *via* the combination of smaller aromatic species [10,79]; however, it is largely derived from indene. H-elimination (R1) and H-abstraction (R2) are the dominant reactions in the consumption of indene, both resulting in the formation of indenyl. As shown in Scheme 1, C–H bond energy of the 1° site is 1.2 kcal/mol lower than its weakest C–C bonds. Unimolecular C–H decomposition should be favored compared to the unimolecular C–C bond fission. Laskin and Lifshitz [62] estimated the rate constant for C–H bond dissociation of indene (R1) to be $1.1 \times 10^{15} e^{-77{,}000/RT}$ cm^3 mol^{-1} s^{-1}, which is pressure-independent. However, different values were adopted in later models [7,8,66,77]. For example, Lifshitz et al. [66] and Pousse et al. [77] also considered R1 to be pressure-independent and temperature-independent; they estimated the rate constant to be 6.0×10^{13} and 1.0×10^{14} cm^3 mol^{-1} s^{-1}, respectively. Blanquart et al. [7] proposed R1 in its reverse direction and recommended its value to be same as that of cyclopentadiene, $1.73 \times 10^{68} T^{-15.16} e^{-116{,}372/RT}$ s^{-1}. Laskin and Lifshitz [62] evaluated the rate constant of R2 as $1.0 \times 10^{14} e^{-5500/RT}$ cm^3 mol^{-1} s^{-1} according to the measurements of cyclopentadiene decomposition by Roy et al. [91]. In some literature models, this rate constant was also proposed in analogy to H-abstraction reactions of other alkenes [3,92,93]. In this study, we adopt the rate constants of R1 and R2 analogously to those of cyclopentadiene. Here, R1 is taken from the H-elimination of cyclopentadiene in Jet-surf 2.0 model [82], which was also adopted in some other PAH models [8,56,77]. The rate constant of R2 is referred to the calculations of Robinson and Lindstedt [34] on cyclopentadiene.

4.1.2. Indenyl (C_9H_7) and indanyl (C_9H_9) decomposition reactions

Indenyl radical is very difficult to decompose at intermediate temperatures (~1200 K) compared to other fuel radicals [62,94,95]. According to the models of Yuan et al. [56], Slavinskaya et al. [39], and Matsugi and Miyoshi [57], three decomposition pathways of indenyl are included in present model, forming cyclopentadienyl + butadiyne, benzyne + propargyl, and fulvenallenyl + acetylene.

H-addition reactions of indene (R3-R4) form two indanyl isomers ($C_9H_9{}^1$: radical on 1° site, $C_9H_9{}^2$: radical on 2° site). The submechanism of indanyl radical was studied by Pousse et al. [77]. Here, we include that part of their model in the present model. Further reactions of two indanyl isomers will be analyzed in the discussion section (Section 5.1.2).

4.2. PAH formation via indenyl reactions

PAH formation mechanism is too complicated to be fully reviewed in this section. We will limit to the specific molecule of this study. C_9H_7 has three isomers, labeled $C_9H_7{}^1$ (indenyl), $C_9H_7{}^2$ and $C_9H_7{}^3$. However, considering the high energy barrier (>100 kcal/mol, see *SMM3*) of H-shift from indenyl to other isomers ($C_9H_7{}^2$ and $C_9H_7{}^3$), indenyl is the most favorable precursor of further PAH growth pathways. Hence, the formation pathways of C_{10}–C_{18} PAHs in this work are considered based on the indenyl radical.

4.2.1. C_{10} aromatic species

The mechanism of C_{10} species describes the conversion from indenyl to naphthalene *via* the addition of a methyl radical [66,78,96]. Reactions of methyl-indene were considered to be rather important pathway in naphthalene formation [53]; therefore, we included all three methyl-indene ($C_{10}H_{10}$) isomers in the present model. The isomerization reactions of $C_{10}H_{10}$ and $C_{10}H_9$ species are very quick [78]. 3-Methyl-indene could thus be a more favorable intermediate due to its lower energy among the three isomers [78]. As shown in Table 1, combination of indenyl and methyl radicals forms 1-mehtyl-indene (R5). It later isomerizes to 2- and 3-methyl-indene. Further decomposition and H-abstraction reactions of methyl-indene isomers produce various $C_{10}H_9$ radicals. Well-skipping reaction between indenyl and methyl may also directly form $C_9H_6{}^3CH_3$ radical (R16). According to the calculations of Dubnikova and Lifshitz [78], only three $C_{10}H_9$ radicals, 3-methyl-1-indenyl ($C_9H_6{}^3CH_3$), 1-indenyl-methyl ($C_9H_7{}^1CH_2$) and 2-indenyl-methyl ($C_9H_7{}^2CH_2$), are considered to be favorable in the subsequent isomerization reactions. Further $C_{10}H_9$ isomerization reactions (R6-R13) yield α- and β-naphthyl radicals ($aC_{10}H_9$ and $bC_{10}H_9$), which are naphthalene ($C_{10}H_8$) precursors (R14 and R15). In addition, ipso-reactions of methyl-indene isomers yield methyl and indene (R17-R19).

4.2.2. C_{11} aromatic species

We considered two groups of species for C_{11} aromatics: ethynyl-indene and methyl-naphthalene. The combination between indenyl and acetylene is proposed to form naphthylmethyl ($A_2{}^1CH_2$) and benzo-cycloheptatrien-5-yl ($cC_{11}H_9$) radicals. Ethynyl-indene and methyl-naphthalene are then formed in a few elementary steps, as shown in Table 1. R20-R26 are analogized to those of cyclopentadienyl [64]; their rate constants are taken from the calculations in literature [64,79,83,84].

4.2.3. C_{12} aromatic species

Two $C_{12}H_8$ isomers are considered in the present model, acenaphthylene (A_2R_5) and ethynyl-naphthalene (A_2C_2H). According to the literature models, acenaphthylene it is either proposed to be formed through the combination of indenyl and propargyl [7,10,97] or the cyclization of naphthyl-2-vinyl radical ($A_2C_2H_2$) [14,16]. A global rate constant used in literature models [8,56] is referred to the estimation of Blanquart et al. [7], which was based on the assumption that indenyl radical gets a benzyl-like structure by its reaction with propargyl (see the part of molecule highlighted in red in Scheme 2). As a result, this reaction consumes two relatively less reactive radicals (C_9H_7 and C_3H_3), but produces two H-atoms that are much more active. Recent calculations of Long

Table 1
Reactions updated in the present model.

No.	Reactions		A^a	n	$E_a{}^a$	Ref.
1	$C_9H_7 + H\ (+M) = C_9H_8\ (+M)$		1.10×10^{14}	0	0	a
	low		4.40×10^{80}	−18.28	12,994	
	troe	0.068	400.7	4135.8	5501.9	
	$H_2/2.0/\ H_2O/6.0/\ CH_4/2.0/\ CO/1.5/\ CO_2/2.0/$					
2	$C_9H_8 + H = C_9H_7 + H_2$		2.22×10^{15}	1.874	4259	b
3	$C_9H_8 + H = C_9H_9{}^1$		6.50×10^{12}	0	3100	[79]
4	$C_9H_8 + H = C_9H_9{}^2$		5.50×10^{12}	0	7000	[79]
5	$C_9H_7 + CH_3 = C_9H_7{}^1CH_3$		8.34×10^{15}	−0.7	−500	[51]
6	$C_9H_6{}^3CH_3 = C_9H_7{}^1CH_2$		2.42×10^{28}	1	52,470	[78]
7	$C_9H_7{}^1CH_2 = C_{10}H_9{}^1$		2.42×10^{28}	1	12,282	[78]
8	$C_9H_7{}^2CH_2 = C_{10}H_9{}^2$		2.42×10^{28}	1	50,520	[78]
9	$C_{10}H_9{}^2 = C_{10}H_9{}^1$		2.42×10^{28}	1	6970	[78]
10	$C_{10}H_9{}^1 = bC_{10}H_9$		2.42×10^{28}	1	15,120	[78]
11	$C_9H_7{}^2CH_2 = C_{10}H_9{}^3$		2.42×10^{28}	1	53,950	[78]
12	$C_{10}H_9{}^3 = C_{10}H_9{}^4$		2.42×10^{28}	1	21,090	[78]
13	$C_{10}H_9{}^4 = aC_{10}H_9$		2.42×10^{28}	1	42,270	[78]
14	$aC_{10}H_9 = C_{10}H_8 + H$		2.42×10^{28}	−5.83	23,947	[80]
15	$bC_{10}H_9 = C_{10}H_8 + H$		2.42×10^{28}	−5.83	23,947	[80]
16	$C_9H_7 + CH_3 = C_9H_6{}^3CH_3 + H$					c
	plog	0.01	8.13×10^{25}	−3.51	41,018	
	plog	1	3.83×10^{26}	−3.69	41,579	
	plog	10	1.16×10^{33}	−5.47	47,110	
	plog	100	1.03×10^{48}	−9.54	61,181	
17	$C_9H_7{}^1CH_3 + H = C_9H_8 + CH_3$		8.00×10^{13}	0	10,000	[66]
18	$C_9H_7{}^2CH_3 + H = C_9H_8 + CH_3$		8.00×10^{13}	0	10,000	[66]
19	$C_9H_7{}^3CH_3 + H = C_9H_8 + CH_3$		8.00×10^{13}	0	10,000	[66]
20	$C_9H_7 + C_2H_2 = C_9H_7C_2H + H$					d
	plog	0.0001	5.58×10^{11}	−0.1	34,720	
	plog	0.001	4.55×10^{7}	1.52	24,370	
	plog	0.01	5.75×10^{6}	1.79	24,100	
	plog	0.1	4.52×10^{7}	1.54	24,710	
	plog	1	8.17×10^{11}	0.33	27,430	
	plog	10	4.40×10^{15}	−0.68	30,890	
	plog	100	7.24×10^{15}	−0.61	34,040	
	plog	1000	1.88×10^{7}	2.07	34,340	
21	$A_2{}^1CH_2 => C_9H_7C_2H + H$					e
	plog	0.013	4.47×10^{129}	−32.57	162,410	
	plog	0.130	1.29×10^{123}	−30.34	163,830	
	plog	1.000	2.95×10^{97}	−22.95	148,280	
	plog	1.500	8.20×10^{14}	0	80,700	
	plog	10.00	1.26×10^{39}	−6.76	99,710	
	plog	50.00	1.44×10^{13}	0.453	76,300	
22	$C_9H_7C_2H + H => A_2{}^1CH_2$		8.80×10^{8}	1.2	−2020	f
23	$A_2{}^1CH_2 = cC_{11}H_9$					g
	plog	0.013	1.12×10^{128}	−32.57	162,410	
	plog	0.130	3.23×10^{121}	−30.34	163,830	
	plog	1.000	7.38×10^{95}	−22.95	148,280	
	plog	1.500	2.05×10^{13}	0	80,700	
	plog	10.00	3.15×10^{37}	−6.76	99,710	
	plog	50.00	3.60×10^{11}	0.453	76,300	
24	$cC_{11}H_9 + H = A_2{}^1CH_2 + H$		2.20×10^{63}	−13	58,628	h
25	$cC_{11}H_9 + H = A_2\text{-} + CH_3$		1.20×10^{61}	−12.4	57,003	i
26	$C_9H_7 + C_2H_2 = cC_{11}H_9$					j
	plog	0.0001	8.28×10^{80}	−22.15	31,790	
	plog	0.0010	3.61×10^{73}	−19.65	30,230	
	plog	0.0100	1.31×10^{64}	−16.55	27,790	
	plog	0.1000	3.69×10^{53}	−13.14	24,810	
	plog	1.0000	1.27×10^{43}	−9.8	21,720	
	plog	10.000	1.07×10^{34}	−6.9	19,000	
	plog	100.00	1.20×10^{27}	−4.67	16,940	
	plog	1000.0	1.09×10^{23}	−3.37	15,770	
27	$C_9H_7 + C_3H_3 = R_5C_{12}H_{10}$		5.78×10^{17}	−1.568	455	k
	duplicate		3.14×10^{19}	−2.163	1195	
28	$C_9H_7 + C_3H_3 = R_5C_{12}H_9 + H$					l
	plog	0.04	1.05×10^{53}	−11.88	28,757	
	plog	1.00	1.70×10^{48}	−9.977	26,755	
	plog	10.0	3.67×10^{26}	−3.879	28,963	
29	$R_5C_{12}H_9 + H\ (+M) = C_9H_7C_3H_3\ (+M)$		1.00×10^{14}	0	0	a
	low		4.40×10^{80}	−18.28	12,994	
	troe	0.068	400.7	4135.8	5501.9	
	$H_2/2.0/\ H_2O/6.0/\ CH_4/2.0/\ CO/1.5/\ CO_2/2.0/$					
30	$R_5C_{12}H_{10} + H = R_5C_{12}H_9 + H_2$		5.60×10^{13}	0	2259	m

(*continued on next page*)

Table 1 (*continued*)

No.	Reactions		A^a	n	$E_a{}^a$	Ref.
31	$R_5C_{12}H_9 = R_5C_{12}H_8 + H$					n
	plog	0.00132	6.28×10^{71}	−18.204	65,711	
	plog	0.0132	2.82×10^{61}	−14.813	62,790	
	plog	0.132	3.77×10^{47}	−10.443	58,072	
	plog	1	7.54×10^{33}	−6.226	52,769	
	plog	10	4.44×10^{21}	−2.491	47,796	
	plog	100	2.86×10^{13}	0.015	44,395	
32	$R_5C_{12}H_9 = A_2R_5 + H$					o
	plog	0.00132	3.68×10^{63}	−15.806	57,410	
	plog	0.0132	6.88×10^{51}	−12.098	53,279	
	plog	0.132	2.37×10^{38}	−7.917	48,317	
	plog	1	9.04×10^{25}	−4.123	43,308	
	plog	10	4.13×10^{16}	−1.278	39,451	
	plog	100	5.55×10^{12}	−0.065	38,028	
33	$R_5C_{12}H_8 + H = A_2R_5 + H$					p
	plog	0.00132	4.94×10^{18}	−1.28	5411	
	plog	0.0132	2.15×10^{22}	−2.28	8429	
	plog	0.132	5.60×10^{26}	−3.47	12,818	
	plog	1	1.66×10^{25}	−2.99	13,691	
	plog	1.312	5.06×10^{25}	−3.12	14,226	
	plog	13.12	2.20×10^{27}	−3.48	19,199	
34	$R_5C_{12}H_8 = A_2R_5$					p
	plog	0.04	5.62×10^{81}	−19.36	121,500	
	plog	1.00	1.45×10^{45}	−8.90	95,999	
	plog	10.0	2.95×10^{31}	−4.97	88,465	
35	$C_9H_7 + C_3H_3 = A_2C_2H_3$					q
	plog	0.04	1.82×10^{73}	−18.14	31,896	
	plog	1.00	3.16×10^{54}	−12.55	22,264	
	plog	10.0	3.88×10^{49}	−11.01	20,320	
36	$C_9H_7 + C_5H_6 = C_9H_7C_5H_6$		1.00×10^{-4}	4.2	3400	r
37	$C_9H_8 + C_5H_5 = C_9H_8C_5H_5$		1.00×10^{-4}	4.2	3400	r
38	$C_9H_7C_5H_6 = fC_{13}H_{10}CH_3$		7.00×10^{9}	1.17	48,500	s
39	$C_9H_7C_5H_6 = bC_{13}H_{10}CH_3$		7.00×10^{9}	1.17	48,500	s
40	$C_9H_8C_5H_5 = fC_{13}H_{10}CH_3$		7.00×10^{9}	1.17	48,500	s
41	$C_9H_8C_5H_5 = bC_{13}H_{10}CH_3$		7.00×10^{9}	1.17	48,500	s
42	$fC_{13}H_{10}CH_3 = C_{13}H_{10} + CH_3$		1.52×10^{13}	0.396	32,820	t
43	$bC_{13}H_{10}CH_3 = bC_{13}H_{10} + CH_3$		1.52×10^{13}	0.396	32,820	t
44	$A_2{}^1CH_2 + C_2H_2 = bC_{13}H_{10} + H$		3.12×10^{-6}	4.712	1417.5	u
45	$A_2 + C_3H_3 = bC_{13}H_{10} + H$		6.26×10^{9}	2.61	56,500	v
46	$C_9H_7 + C_5H_5 = C_9H_7C_5H_5$					w
	plog	0.04	3.14×10^{174}	−49.43	62,602	
	plog	1.00	1.80×10^{188}	−52.62	76,380	
	plog	10.0	4.28×10^{174}	−47.91	75,125	
47	$C_9H_7 + C_5H_5 = C_9H_6C_5H_5 + H$		5.60×10^{13}	0	2259	x
48	$C_9H_6C_5H_5 + H\ (+M) = C_9H_7C_5H_5\ (+M)$		1.00×10^{14}	0	0	a
	low		4.40×10^{80}	−18.28	12,994	
	troe	0.068	400.7	4135.8	5501.9	
	H_2/2.0/ H_2O/6.0/ CH_4/2.0/ CO/1.5/ CO_2/2.0/					
49	$C_9H_7C_5H_5 + H = C_9H_6C_5H_5 + H_2$		1.12×10^{14}	0	2259	m
50	$C_9H_6C_5H_5 = A_3H$		5.60×10^{8}	1.31	19,100	y
51	$A_3H = A_3 + H$		6.80×10^{11}	0.508	18,659	z
52	$C_9H_7C_5H_6 = C_9H_7C_5H_5 + H$		1.55×10^{10}	1.16	45,500	aa
53	$C_9H_8C_5H_5 = C_9H_7C_5H_5{}^b + H$		1.55×10^{10}	1.16	45,500	aa
54	$fC_{13}H_9 + CH_3 = fC_{13}H_9CH_3$		6.62×10^{13}	−0.7	−500	bb
55	$bC_{13}H_9 + CH_3 = bC_{13}H_9CH_3$		6.62×10^{13}	−0.7	−500	bb
56	$A_3CH_2 = bC_{13}H_9 + C_2H_2$					cc
	plog	0.013	4.08×10^{134}	−34.08	169,130	
	plog	0.130	1.29×10^{112}	−27.5	155,360	
	plog	1.000	1.10×10^{89}	−20.82	141,680	
	plog	10.00	1.03×10^{55}	−11.23	116,580	
	plog	50.00	1.44×10^{13}	0.453	76,300	
57	$A_3CH_2 = cC_{15}H_{11}$					g
	plog	0.013	1.12×10^{128}	−32.57	162,410	
	plog	0.13	3.23×10^{121}	−30.34	163,830	
	plog	1	7.38×10^{95}	−22.95	148,280	
	plog	1.5	2.05×10^{13}	0	80,700	
	plog	10	3.15×10^{37}	−6.76	99,710	
	plog	50	3.60×10^{12}	0.453	76,300	
58	$C_9H_7 + C_6H_5 = C_9H_7C_6H_5$					w
	plog	0.013	3.14×10^{174}	−49.43	62,602	
	plog	1.000	1.90×10^{188}	−52.62	76,380	
	plog	100.0	4.30×10^{174}	−47.91	75,125	

(*continued on next page*)

Table 1 (*continued*)

No.	Reactions	A^a	n	$E_a{}^a$	Ref.
59	$C_9H_6C_6H_5 + H\ (+M) = C_9H_7C_6H_5\ (+M)$	1.00×10^{14}	0	0	a
	low	4.40×10^{80}	−18.28	12,994	
	troe 0.068	400.7	4135.8	5501.9	
	$H_2/2.0/\ H_2O/6.0/\ CH_4/2.0/\ CO/1.5/\ CO_2/2.0/$				
60	$C_9H_7C_6H_5 + H = C_9H_6C_6H_5 + H_2$	1.12×10^{14}	0	2259	m
61	$C_9H_6C_6H_5 = cC_{15}H_{11}$	8.40×10^{11}	0.456	42,368	[47]
62	$cC_{15}H_{11} = cpC_{15}H_{10} + H$	6.40×10^{12}	0	34,800	dd
63	$C_9H_7 + A_1CH_2 = C_9H_7C_7H_7$	8.10×10^{11}	−0.369	1172	[47]
64	$C_9H_7C_7H_7 = C_9H_7CHC_6H_5 + H$	1.30×10^{11}	0.77	4354	[47]
65	$C_9H_7C_7H_7 + H = C_9H_7CHC_6H_5 + H_2$	4.32	3.98	1380	[47]
66	$C_9H_7CHC_6H_5 = C_{10}H_8C_6H_5$	8.10×10^{11}	0.038	11,810	[47]
67	$C_{10}H_7C_6H_5 + H = C_{10}H_8C_6H_5$				ee
	plog 0.0263	1.16×10^{41}	−9.51	10,830	
	plog 0.1184	2.40×10^{40}	−9.06	11,570	
	plog 1.0000	6.82×10^{35}	−7.37	11,230	
68	$C_{10}H_6C_6H_5 + H\ (+M) = C_{10}H_7C_6H_5\ (+M)$	1.00×10^{14}	0	0	ff
	low	$6.6. \times 10^{75}$	−16.3	7000	
	troe 1	0.1	584.9	6113	
	$H_2/2.0/\ H_2O/6.0/\ CH_4/2.0/\ CO/1.5/\ CO_2/2.0/\ C_2H_6/3.0/$				
69	$C_{10}H_7C_6H_5 + H = C_{10}H_6C_6H_5 + H_2$	5.90×10^{11}	0.705	12,626	[47]
70	$C_{10}H_6C_6H_5 = C_{16}H_{11}{}^f$	8.40×10^{11}	0.456	42,368	[47]
71	$C_{16}H_{11}{}^f = C_{16}H_{11}{}^p$	6.60×10^{11}	0.566	39,778	[47]
72	$C_{16}H_{11}{}^f = fC_{16}H_{10} + H$	6.90×10^{11}	0.548	25,832	[47]
73	$C_{16}H_{11}{}^p = pA_4 + H$	6.80×10^{11}	0.508	18,659	[47]
74	$cpC_{15}H_9 + CH_3 = cpC_{15}H_9CH_3$	6.62×10^{13}	−0.7	−500	bb
75	$cpC_{15}H_9 + CH_3 = cpC_{15}H_8CH_3 + H$				c
	plog 0.01	3.03×10^{15}	−0.58	10,871	
	plog 0.10	1.23×10^{31}	−1.4	24,769	
	plog 1.00	3.23×10^{46}	−5.9	39,791	
	plog 10.0	1.49×10^{56}	−9.05	52,592	
76	$cpC_{15}H_9CH_3 = cpC_{15}H_8CH_3 + H$				gg
	plog 0.10	1.74×10^{97}	−24.38	116,128	
	plog 1.00	5.77×10^{97}	−24.0	122,296	
	plog 10.0	1.75×10^{87}	−20.6	120,401	
77	$cpC_{15}H_9CH_3 + H = cpC_{15}H_8CH_3 + H_2$	1.30×10^{6}	2.38	2800	[81]
78	$cpC_{15}H_8CH_3 = pA_4 + H$	9.25×10^{43}	−9.191	55,757	hh
	duplicate	1.98×10^{31}	−6.066	20,630	
79	$C_9H_7 + C_9H_8 = C_9H_8C_9H_7$	1.00×10^{-4}	4.2	3400	r
80	$C_9H_8C_9H_7 = C_{17}H_{12}CH_3{}^b$	7.00×10^{9}	1.17	48,500	s
81	$C_{17}H_{12}CH_3{}^b = BCF + CH_3$	1.52×10^{13}	0.396	32,820	t
82	$bC_{13}H_9 + C_5H_6 = bC_{13}H_9C_5H_6$	1.00×10^{-4}	4.2	3400	r
83	$bC_{13}H_9C_5H_6 = C_{17}H_{12}CH_3{}^b$	7.00×10^{9}	1.17	48,500	s
84	$bC_{13}H_{10} + C_5H_5 = bC_{13}H_{10}C_5H_5$	1.00×10^{-4}	4.2	3400	r
85	$bC_{13}H_{10}C_5H_5 = C_{17}H_{12}CH_3{}^b$	7.00×10^{9}	1.17	48,500	s
86	$C_{17}H_{12}CH_3{}^b = BCF^5CH_3 + H$	6.80×10^{11}	0.508	18,659	z
87	$BCF\text{-} + CH_3 = BCF^{11}CH_3$	6.62×10^{13}	−0.7	−500	bb
88	$C_9H_7 + C_9H_7 \Rightarrow C_9H_7C_9H_7$				w
	plog 0.0132	3.14×10^{174}	−49.43	62,602	
	plog 1.0000	1.90×10^{188}	−52.62	76,380	
	plog 100.00	4.30×10^{174}	−47.91	75,125	
89	$C_9H_7C_9H_7 \Rightarrow C_9H_7 + C_9H_7$				w
	plog 0.0132	2.08×10^{70}	−17.55	62,254	
	plog 1.0000	1.47×10^{77}	−18.71	73,651	
	plog 100.00	1.24×10^{64}	−14.25	72,842	
90	$bC_{13}H_9 + C_5H_5 = bC_{13}H_9C_5H_5$				w
	plog 0.0132	3.14×10^{174}	−49.43	62,602	
	plog 1.0000	1.90×10^{188}	−52.62	76,380	
	plog 100.00	4.30×10^{174}	−47.91	75,125	
91	$fC_{13}H_9 + C_5H_5 = fC_{13}H_9C_5H_5$				w
	plog 0.0132	3.14×10^{174}	−49.43	62,602	
	plog 1.0000	1.90×10^{188}	−52.62	76,380	
	plog 100.00	4.30×10^{174}	−47.91	75,125	
92	$C_9H_6C_9H_7 + H\ (+M) = C_9H_7C_9H_7\ (+M)$	1.00×10^{14}	0	0	a
	low	4.40×10^{80}	−18.28	12,994	
	troe 0.068	400.7	4135.8	5501.9	
	$H_2/2.0/\ H_2O/6.0/\ CH_4/2.0/\ CO/1.5/\ CO_2/2.0/$				
93	$C_9H_7C_9H_7 + H = C_9H_6C_9H_7 + H_2$	1.12×10^{14}	0	2259	[57]
94	$C_9H_7C_9H_7 + C_9H_7 = C_9H_6C_9H_7 + C_9H_8$	3.20×10^{11}	0	15,100	ii
95	$C_9H_6C_9H_7 = C_{18}H_{13}{}^c$	5.60×10^{8}	1.31	19,100	y
96	$C_9H_6C_9H_7 = C_{18}H_{13}{}^p$	5.60×10^{8}	1.31	19,100	y
97	$C_{18}H_{13}{}^c = C_{18}H_{12}{}^c + H$				z
	plog 0.04	3.40×10^{10}	0.78	30,230	
	plog 1.00	6.80×10^{11}	0.508	18,659	

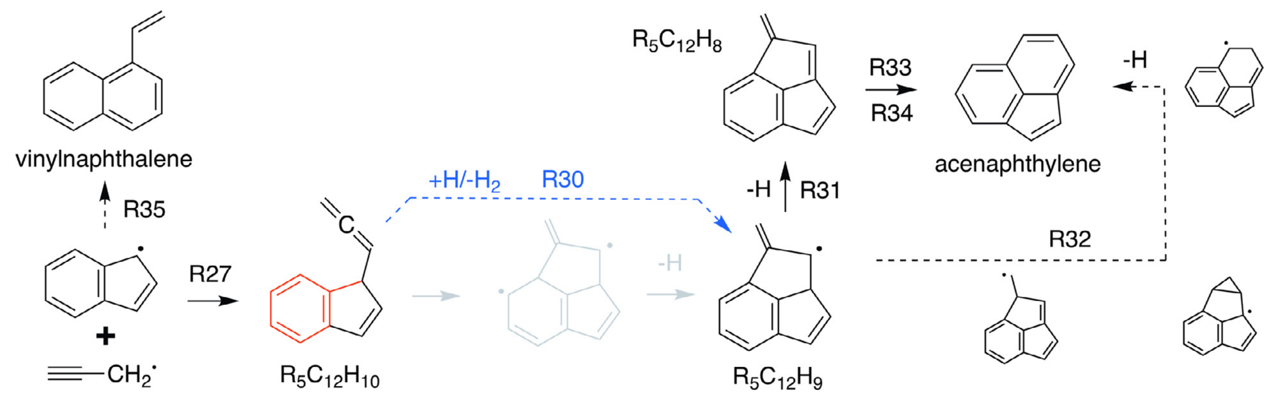
(*continued on next page*)

Table 1 (*continued*)

No.	Reactions		A^a	n	$E_a{}^a$	Ref.
98	$C_{18}H_{13}{}^p = C_{18}H_{12}{}^p + H$					z
	plog	0.04	3.40×10^{10}	0.78	30,230	
	plog	1.00	6.80×10^{11}	0.508	18,659	
99	$C_9H_7 + C_9H_7 = C_9H_6C_9H_7 + H$		5.25×10^{14}	−0.853	7254	x
100	$bC_{13}H_8C_5H_5 + H\ (+M) = bC_{13}H_9C_5H_5\ (+M)$		1.00×10^{14}	0	0	a
	low		4.40×10^{80}	−18.28	12,994	
	troe	0.068	400.7	4135.8	5501.9	
	H_2/2.0/ H_2O/6.0/ CH_4/2.0/ CO/1.5/ CO_2/2.0/					
101	$bC_{13}H_9C_5H_5 + H = bC_{13}H_8C_5H_5 + H_2$		5.60×10^{13}	0	2259	m
102	$bC_{13}H_8C_5H_5 = C_{18}H_{13}{}^c$		5.60×10^{8}	1.31	19,100	y
103	$bC_{13}H_8C_5H_5 = C_{18}H_{13}{}^p$		5.60×10^{8}	1.31	19,100	y
104	$bC_{13}H_9 + C_5H_5 = bC_{13}H_8C_5H_5 + H$		5.25×10^{14}	−0.853	7254	x
	duplicate		2.00×10^{10}	0.951	15,793	
105	$fC_{13}H_8C_5H_5 + H\ (+M) = fC_{13}H_9C_5H_5\ (+M)$		1.00×10^{14}	0	0	a
	low		4.40×10^{80}	−18.28	12,994	
	troe	0.068	400.7	4135.8	5501.9	
	H_2/2.0/ H_2O/6.0/ CH_4/2.0/ CO/1.5/ CO_2/2.0/					
106	$fC_{13}H_9C_5H_5 + H = fC_{13}H_8C_5H_5 + H_2$		5.60×10^{13}	0	2259	m
107	$fC_{13}H_8C_5H_5 = C_{18}H_{13}{}^t$		5.60×10^{8}	1.31	19,100	y
108	$C_{18}H_{13}{}^t = C_{18}H_{12}{}^t + H$		6.80×10^{11}	0.508	18,659	z
109	$fC_{13}H_9 + C_5H_5 = fC_{13}H_8C_5H_5 + H$		5.25×10^{14}	−0.853	7254	x
	duplicate		2.00×10^{10}	0.951	15,793	

[a]The rate constants are given in the form of $k = A \cdot T^n \cdot e^{-E_a/RT}$, where A has the units of cm, mol and s; T has the unit of K; E_a has the unit of cal/mol.

a Rate constant analogized to $C_5H_5 + H = C_5H_6$ in JetSurf 2.0 model [82].
b Rate constant analogized to $C_5H_6 + H = C_5H_5 + H_2$, calculated by Robinson and Lindstedt [34].
c Rate constant analogized to $C_5H_5 + CH_3 = C_5H_5CH_2 + H$, calculated by Sharma and Green [51].
d Rate constant analogized to $C_5H_5 + C_2H_2 = C_7H_6 + H$ calculated by Da Silva et al. [83].
e Rate constant analogized to $A_1CH_2 = C_7H_6 + H$ calculated by Polino and Cavallotti [84].
f Rate constant analogized to $C_7H_6 + H = A_1CH_2$ calculated by Cavallotti et al. [85].
g Rate constant analogized to $A_1CH_2 = cC_7H_7$ estimated in the model of Yuan et al. [56].
h Rate constant analogized to $cC_7H_7 + H = A_1CH_2 + H$ calculated by Fascella et al. [79].
i Rate constant analogized to $cC_7H_7 + H = A_1\text{-} + CH_3$ calculated by Fascella et al. [79].
j Rate constant analogized to $C_5H_5 + C_2H_2 = cC_7H_7$ calculated by Da Silva et al. [83].
k Rate constant taken equal to $A_1CH_2 + C_3H_3 = A_1CH_2C_3H_3$ calculated by Matsugi and Miyoshi [45].
l Rate constant taken equal to $A_1CH_2 + C_3H_3 = C_{10}H_9 + H$ in the model of Matsugi and Miyoshi [57].
m Rate constant analogized to $C_5H_5C_5H_5 + H = C_5H_5C_5H_4 + H_2$ calculated Matsugi and Miyoshi [57].
n Rate constant taken equal to $C_9H_6CH_3 = C_9H_6CH_2 + H$ calculated by Matsugi and Miyoshi [45].
o Rate constant taken equal to $C_9H_6CH_3 = A_2 + H$ calculated by Matsugi and Miyoshi [45].
p Rate constant taken equal to the isomerization of $C_9H_6CH_2$ and A_2 calculated by Matsugi and Miyoshi [45].
q Rate constant analogized to $C_5H_5 + C_3H_3 = A_1C_2H_3$ in the model of Miller and Klippenstein [42] and Yuan et al. [55].
r Rate constant analogized to $C_5H_5 + C_5H_6 = C_5H_6C_5H_5$ calculated by Kislov and Mebel [86].
s Rate constant taken equal to the controlling step of the isomerization from $C_5H_6C_5H_5$ to $C_9H_6CH_3$, calculated by Kislov and Mebel [86].
t Rate constant taken equal to $C_6H_{10}CH_3 = C_6H_{10} + CH_3$ in the model of Narayanaswamy et al. [87].
u Rate constant analogized to $A_1CH_2 + C_2H_2 = C_9H_8 + H$ by the calculation of Vereecken and Peeters [88].
v Product of this reaction changes to benzo-indene, it is proposed to form fluorene in the model of Yuan et al. [56].
w Rate constant analogized to $C_5H_5 + C_5H_5 = C_5H_5C_5H_5$ calculated by Matsugi and Miyoshi [57].
x Rate constant analogized to $C_5H_5 + C_5H_5 = C_5H_5C_5H_4 + H$ calculated by Cavallotti and Polino [46].
y Rate constant analogized to $C_5H_5C_5H_4 = C_{10}H_9$ calculated by Kislov and Mebel [43].
z Rate constant analogized to $C_{16}H_{11} = pA_4 + H$ calculated by Sinha et al. [47].
aa Rate constant analogized to $C_5H_6C_5H_5 = C_5H_5C_5H_5 + H$ calculated by Wang et al. [49].
bb Rate constant analogized to $C_5H_5 + CH_3 = C_5H_5CH_3$ calculated by Sharma and Green [51].
cc Rate constant analogized to $A_1CH_2 = C_5H_5 + C_2H_2$ estimated in the model of Yuan et al. [56].
dd Rate constant analogized to $C_5H_7 = C_5H_6 + H$ estimated in the model of Gueniche et al. [89].
ee Rate constant analogized to $C_6H_5C_6H_5 + H = C_6H_6C_6H_5$ estimated in the model of Wang and Frenklach [14].
ff Rate constant analogized to $A_1\text{-} + H = A_1$ estimated in the model of Wang and Frenklach [14].
gg Rate constant analogized to $C_5H_5CH_3 = C_5H_5CH_2 + H$ calculated by Sharma and Green [51].
hh Rate constant analogized to $C_5H_5CH_2 = A_1 + H$ in the model of Harper et al. [90].
ii Rate constant analogized to $C_5H_5C_5H_5 + C_5H_5 = C_5H_5C_5H_4 + C_5H_6$ calculated by Matsugi and Miyoshi [57].



Scheme 2. Reaction scheme proposed for $C_9H_7 + C_3H_3$. Solid arrows represent elementary reaction steps, dashed arrows represent multiple reaction steps. Blue dashed arrow highlights the key H-abstraction reaction step. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

et al. [48] and Sinha et al. [47] showed that a necessary H-loss step from |R'-R| to |R'-R|* radical (R and R' are functional groups, * is a radical label) should be considered in the model. This approach will effectively adjust the over-estimation of the system reactivity by the global reaction, since it consumes an extra radical in H-abstraction reaction of |R'-R| (three radicals in and one out). Therefore, in this work, we proposed a multi-step scheme for the addition of propargyl to indenyl (R27-34) and analogized to that of benzyl [9,45,56,57] but also included a necessary H-abstraction reaction step (R30). Detailed reaction process is shown in Scheme 2.

The addition of acetylene on naphthyl radical can produce naphthyl-2-vinyl radical. It either cyclizes to acenaphthylene or forms ethynyl-naphthalene *via* C–H bond cleavage. Herein, we propose another vinyl-naphthalene formation reaction *via* the combination of indenyl and propargyl radical (R35), competing with the route proposed for acenaphthylene. This is analogized to the styrene formation reaction by cyclopentadienyl and propargyl [10,40,98].

4.2.4. C_{13} aromatic species

Benz[g]indene, benz[f]indene and fluorene are typical $C_{13}H_{10}$ PAHs that can be formed *via* the combination of indenyl and cyclopentadienyl radicals [50]. H-elimination reactions of these molecules yield benz[g]indenyl, benz[f]indenyl and fluorenyl radicals, which are also cyclopentadienyl-like radicals but have tricyclic rings. We proposed two molecular structure groups for $C_{13}H_{10}$ and $C_{13}H_9$ species in the present model: one group represents the benz-indene isomers ($bC_{13}H_{10}$) and their radicals ($bC_{13}H_9$) while the other group represents fluorene ($fC_{13}H_{10}$) and fluorenyl radical ($fC_{13}H_9$). Benz-indenyl is treated separately from fluorenyl because the chemical reactivity of the five-member ring in this molecule is impacted by one benzene ring on its single side whilst that of fluorenyl is impacted on its both sides. Therefore, we consider benz-indenyl to be more active in PAH growth. In addition, fluorenyl and benz-indenyl radicals produce very different large PAHs through subsequent reactions with same intermediates. However, benz[g]indenyl and benz[f]indenyl form similar products in further reactions, are thus lumped.

The combination of indenyl and cyclopentadienyl [50] could be similar to that of two cyclopentadienyls [43,48]. However, the reactions of $C_9H_7 + C_5H_6$ and $C_9H_8 + C_5H_5$ are slightly different from $C_5H_5 + C_5H_6$ because of the non-symmetrical structures of $C_9H_7C_5H_6$ and $C_9H_8C_5H_5$ (see their structures in Scheme 3). These will either eliminate an H-atom forming $C_9H_7C_5H_5$ or undergo isomerization steps forming $fC_{13}H_{10}CH_3$ and $bC_{13}H_{10}CH_3$ radicals. The products of these two channels are similar, mainly fluorene and benzo-indene. We propose the following scheme (R36-43) based on the calculations of Mulholland et al. [50]; rate constants are analogized to the similar reactions between cyclopentadienyl and cyclopentadiene [86].

In addition to the reactions suggested by Mulholland et al. [50], we also propose a formation pathway of benz-indene from naphthylmethyl (R44), similar to the formation of indene from the combination of benzyl and acetylene [88]. For the combination of propargyl and naphthalene (R45) [19,56], we recommend its product to be benz-indene instead of fluorene because the addition of propargyl is more likely to create a fused five-member ring on one side than in the middle of naphthalene.

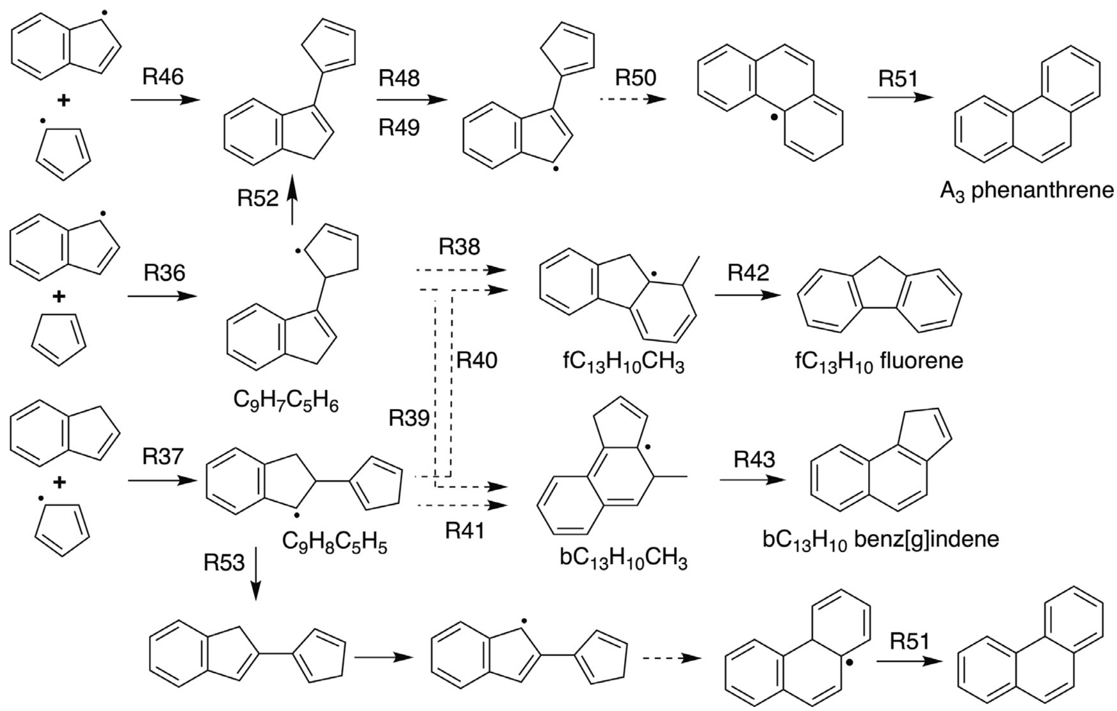
4.2.5. C_{14} aromatic species

Anthracene and phenanthrene are the $C_{14}H_{10}$ isomers identified in previous studies, and are typical tricyclic aromatic species. In our model, we lumped these two isomers, since they have very similar growth pathways and yield products with similar structures. Beside HACA formation pathways for $C_{14}H_{10}$ PAHs [3,14], we propose the combination of indenyl and cyclopentadienyl (R46-51) to be an efficient route [50] (see Scheme 3). The rate constants of R46-51 are again analogized to that of cyclopentadienyl (see Table 1).

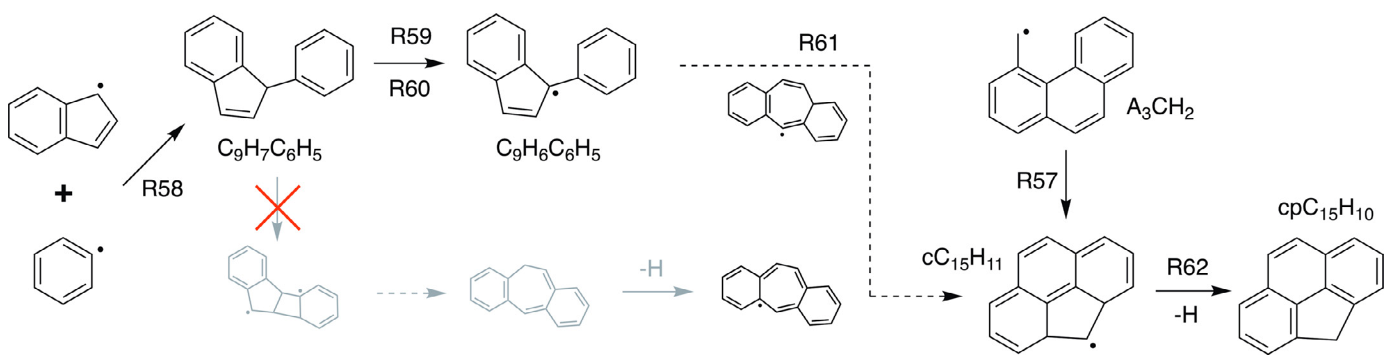
For $C_{14}H_{12}$ species, Yuan et al. [56] and Matsugi and Miyoushi [57] proposed it to be stilbene in their toluene studies. However, benzyl radical concentration is much less than indenyl in the present study; stilbene is thus not that important. Therefore, we assumed $C_{14}H_{12}$ to be methyl-benzindene or methyl-fluorene formed through the combination of methyl and benz-indenyl or fluorenyl (R54-55). Further reactions of both species can form phenanthrene, similar to the transformation from methyl-cyclopentadiene to benzene [51].

4.2.6. C_{15} aromatic species

Phenanthrenyl-methyl radical (A_3CH_2) can be produced by the combination of benz-indenyl and acetylene (R52), which is similar to the conversion from cyclopentadienyl to benzyl. Cyclization of phenanthrenyl-methyl radical yields another radical (R57), see $cC_{15}H_{11}$ in Scheme 4. Thereafter, 4*H*-cyclopenta[*def*]phenanthrene ($cpC_{15}H_{10}$) can be formed *via* β-scission of its C–H bond (R62). We also proposed $cpC_{15}H_{10}$ to be formed through the combination of indenyl and phenyl. As shown in Scheme 4, 1-phenyl-1*H*-indene ($C_9H_7C_6H_5$) loses an H-atom to form $C_9H_6C_6H_5$ radical. It then isomerizes to $cC_{15}H_{11}$ radical via seven-member-ring structure intermediates. A brief reaction scheme (R58-61) is shown in Table 1 with estimated rate constants. Opening of double bonds in $C_9H_7C_6H_5$ and further cyclization is considered negligible, as shown by the red cross in Scheme 4.



Scheme 3. Reaction scheme proposed for $C_9H_8 + C_5H_5$ and $C_9H_7 + C_5H_6$. Solid arrows represent elementary reaction steps; dashed arrows represent reactions with multiple steps.



Scheme 4. Reaction scheme proposed for $C_9H_7 + C_6H_5$. Solid arrows represent elementary reaction steps; dashed arrows represent reactions with multiple steps.

4.2.7. C_{16} aromatic species

We added the reaction of indenyl and benzyl in the present model (R63-R73 in Table 1) for pyrene (pA_4) and fluoranthene ($fC_{16}H_{10}$), which are adopted from the quantum chemical calculations of Sinha et al. [47]. After the combination of indenyl and benzyl (R63), the product $C_9H_7C_7H_7$ undergoes several reaction steps (R64-R67) to form phenyl-naphthalene ($C_{10}H_7C_6H_5$). The following reaction sequence of $C_{10}H_7C_6H_5$ (R68-R73) is the same as the reaction pathway of naphthyl and phenyl combination. The addition reaction of methyl to $cpC_{15}H_9$ is also considered a potential pyrene formation pathway (R74-R78). Besides these routes, we propose the addition of propargyl on fluorenyl to yield fluoranthene, analogized to the addition of propargyl on benz-indenyl [97].

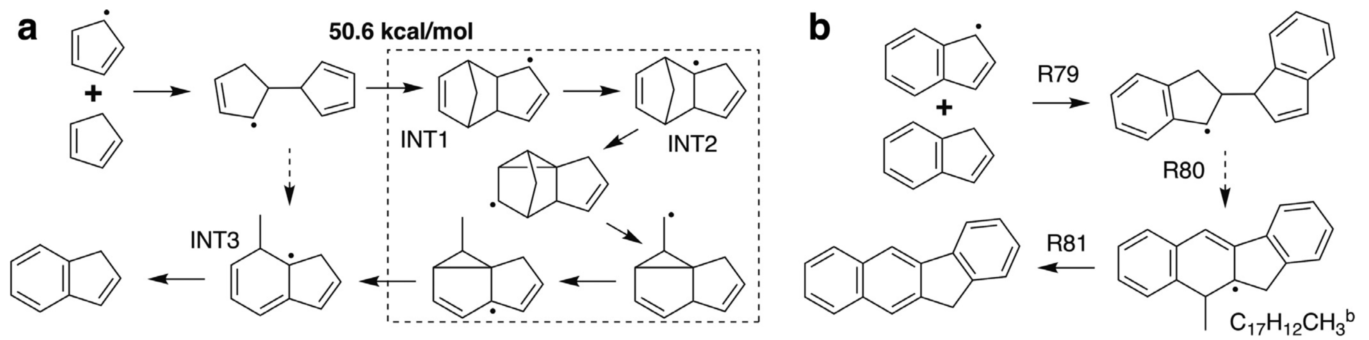
4.2.8. C_{17} aromatic species

C_{17} PAH species have not been detailed in previous studies [11,99,100]; particularly their specific formation pathways and carbon growth reactions are not described adequately. Previous quantum chemical calculations pointed out that the combination of indenyl and indene can form C_{17} and C_{18} species [65]. Three benzofluorene isomers may be formed in this reaction scheme, benzo[*b*]fluorene (BCF) is emphasized as a major isomer [65]. As shown in Scheme 5(a), Kislov et al. [86] calculated a series of $C_{10}H_{11}$ isomerization reactions from 3a,4,7,7a-tetrahydro-1*H*-4,7-methano-indenyl (INT1) to 7-methyl-7,7a-dihydro-1*H*-7a-indenyl (INT3) after the combination of cyclopentadienyl and cyclopentadiene. β-Scission of INT3 preferentially produces indene rather than methyl-indene because of the weaker dissociation energy of C–C bond than C–H bond. A modified scheme for $C_9H_8 + C_9H_7$ (R79-R81 in Table 1) is adopted in the present model analogously to $C_5H_6 + C_5H_5$ reactions, as shown in Scheme 5(b).

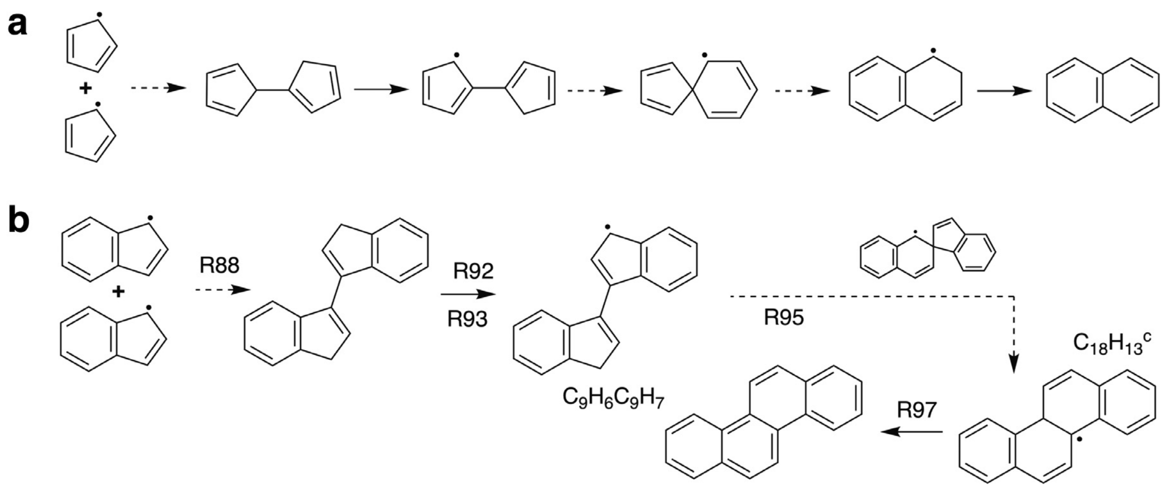
Benzofluorene pathways *via* the combination of benz-indene and cyclopentadienyl or benz-indenyl and cyclopentadiene (R82-R85) are also considered in the present model, following the repetitive chemical reactions of cyclopentadienyl and indenyl (Scheme 5).

4.2.9. C_{18} aromatic species

We consider three kinds of C_{18} PAH species in the present model, with formulae of $C_{18}H_{14}$, $C_{18}H_{12}$ and $C_{18}H_{10}$. $C_{18}H_{14}$ could be 5-methyl-benzofluorene (BCF^5CH_3), formed by the β-scission of $C_{17}H_{12}CH_3{}^b$ (R86) [65]. Another methyl-benzofluorene isomer, labeled as 11-methyl-benzofluorene ($BCF^{11}CH_3$), is the product of the addition of methyl and benzo-fluorenyl radicals (R87). Bi-indene



Scheme 5. Reaction schemes (a) for $C_5H_6 + C_5H_5$ calculated by Kislov and Mebel [86], (b) proposed here for $C_9H_8 + C_9H_7$. Solid arrows represent elementary reaction steps; broad arrows represent reactions with multiple steps.



Scheme 6. Reaction schemes (a) for $C_5H_5 + C_5H_5$, (b) proposed for $C_9H_7 + C_9H_7$. Solid arrows represent elementary reaction steps; dash arrows represent reactions with multiple steps.

isomers are dominant $C_{18}H_{14}$ species [63,65]. Bi-indene has several isomers, but we lumped all isomers as 3,3-bi-indene in order to simplify the present model because they have very similar further reactions [48]. This is the same strategy adopted in the model of Yuan et al. for cyclopentadienyl combination reactions [56]. The related reactions of indenyl combination (R88 and R89 in Table 1) are analogized to those of cyclopentadienyl, as shown in Scheme 6. Furthermore, the combinations of $C_{13}H_9$ isomers and cyclopentadienyl produce other $C_{18}H_{14}$ isomers (R90 and R91), fluorenyl-cyclopentadiene ($fC_{13}H_9C_5H_5$) and benzindenyl-cyclopentadiene ($bC_{13}H_9C_5H_5$).

Three $C_{18}H_{12}$ isomers, chrysene ($C_{18}H_{12}{}^c$), benzo[c]phenanthrene ($C_{18}H_{12}{}^p$), and triphenylene ($C_{18}H_{12}{}^t$), are considered in the present model. Among these, chrysene is a typical $C_{18}H_{12}$ PAH that has been frequently studied in previous models [3,7,14,15,18,19,39,55,56]. $C_9H_7C_9H_7$ eliminates an H-atom yielding $C_9H_6C_9H_7$ radical, and it isomerizes to hydrochrysenyl ($C_{18}H_{13}{}^c$, R95) or hydrobenzo[c]phenanthrenyl ($C_{18}H_{13}{}^p$, R96). The β-scission reactions of $C_{18}H_{13}{}^c$ and $C_{18}H_{13}{}^p$ form chrysene and benzo[c]phenanthrene (R97 and R98), respectively. Another formation channel of $C_9H_6C_9H_7$ radical is the well-skipping reaction of two indenyl radicals (R99). The proposed formation pathway for chrysene is shown in Scheme 6(b), while similar reactions for benzo[c]phenanthrene are omitted.

Subsequent reactions of benzindenyl-cyclopentadiene ($bC_{13}H_9C_5H_5$) may also form chrysene and benzo[c]phenanthrene (R100-R104). Slightly different reactions of fluorenyl and cyclopentadienyl yields triphenylene (R105-R109).

4.2.10. PAHs beyond 18 C-atoms

Since these species are not directly observed in the current experiments, we have provided detailed information of this part in Section 3 of *SMM2*. Simple reaction schemes are included for some C_{19} and C_{20} species, as collected in Table S2. Self-combination of large resonantly stabilized radicals were considered for larger PAHs in previous studies [59,61]. In the present model, global reactions are applied for these pathways.

4.3. Thermodynamic and transport data

Thermodynamic and transport data of C_0–C_6 species used in this work are taken from literature models [56,70], and those of aromatic species are taken from references [7,14,57,97,101]. Thermal data of new species added in the present model are calculated by group-additivity method [102–104].

5. Model validation

In this section, the present model is validated against the current flow reactor experimental data and the shock tube data of Laskin and Lifshitz [62]. Numerical simulations of flow reactor and shock tube was performed with OpenSMOKE++ [105]. Energy equation was not solved in the flow reactor; instead temperature profiles measured in the experiment were used as input parameters (see *SMM1*). Shock tube was modeled as an adiabatic batch reactor, and the residence time was set to 1.7 ms according to the experimental method reported in [106]. The main dissociation

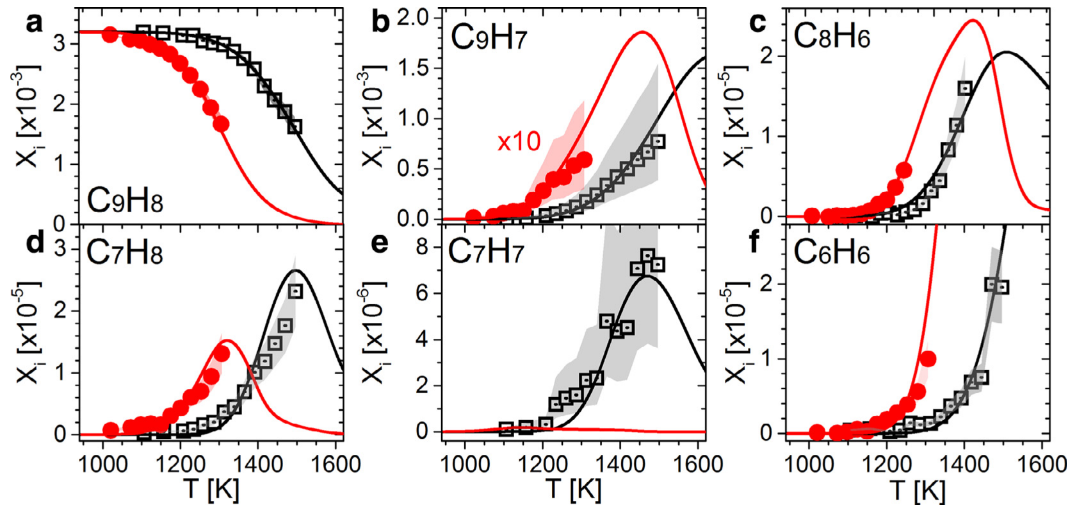


Fig. 1. Measured mole fraction profiles of indene (C_9H_8) and its main decomposition products in a flow reactor at 30 (*black squares*) and 760 (*red circles*) Torr. Shadows with the data plots represent the uncertainty of the measurements. Lines are predictions of the present model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

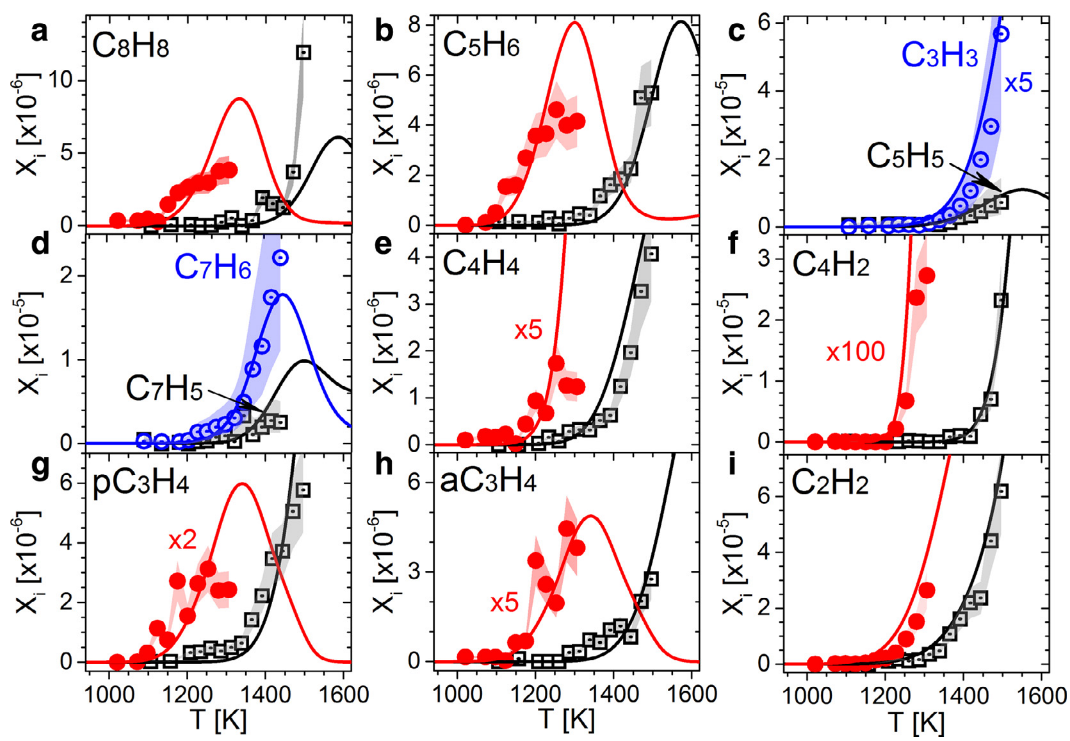


Fig. 2. Mole fraction profiles of small intermediates measured in indene pyrolysis in a flow reactor at 30 (*black squares*) and 760 (*red circles*) Torr. The blue circles in (c) and (d) are the mole fractions of propargyl and 5-vinylidene-1,3-cyclopentadiene at 30 Torr. Shadows with the data plots represent the uncertainty of the measurements. Lines are the predictions of the present model. Experimental data and simulations of species with small concentration are scaled with a common factor. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

channels of indene and ring expansion routes from indenyl radical are illustrated based on the kinetic analysis of the present model.

5.1. Indene pyrolysis in flow reactor

Dozens of pyrolysis products were detected in the flow reactor pyrolysis experiments of indene, and their mole fraction profiles are plotted as a function of reaction temperature. The experimental and simulated mole fraction profiles of pyrolysis species are shown in Figs. 1–4. In general, the current model can adequately capture the decomposition or formation temperature windows of the observed pyrolysis species, and the quantitative agreement between the experiments and model is quite reasonable. Rate of production (ROP) analysis and sensitivity analysis are used to provide insights into the chemistry of indene decomposition and aromatics growth.

5.1.1. Primary decomposition of indene

Indene consumption profiles are presented in Fig. 1(a) and compared with simulation results. Only 50% of indene conversion is observed during the experiments at ~1300 and 1500 K for 30 and 760 Torr, respectively. Higher conversions of indene at higher temperatures were not measured due to the increasing influence of soot on the sampling process. The present model captured the temperature dependencies of indene consumption at both pres-

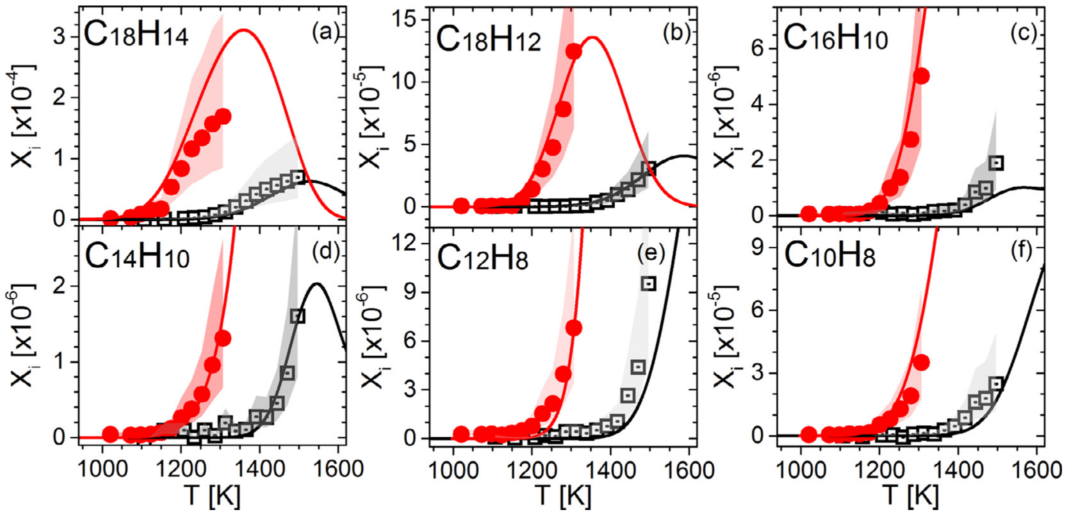


Fig. 3. Mole fraction profiles of dominant (common) PAHs measured in indene pyrolysis in a flow reactor at 30 (*black squares*) and 760 (*red circles*) Torr. Shadows with the data plots represent the uncertainty of the measurements. Lines indicate the model predictions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

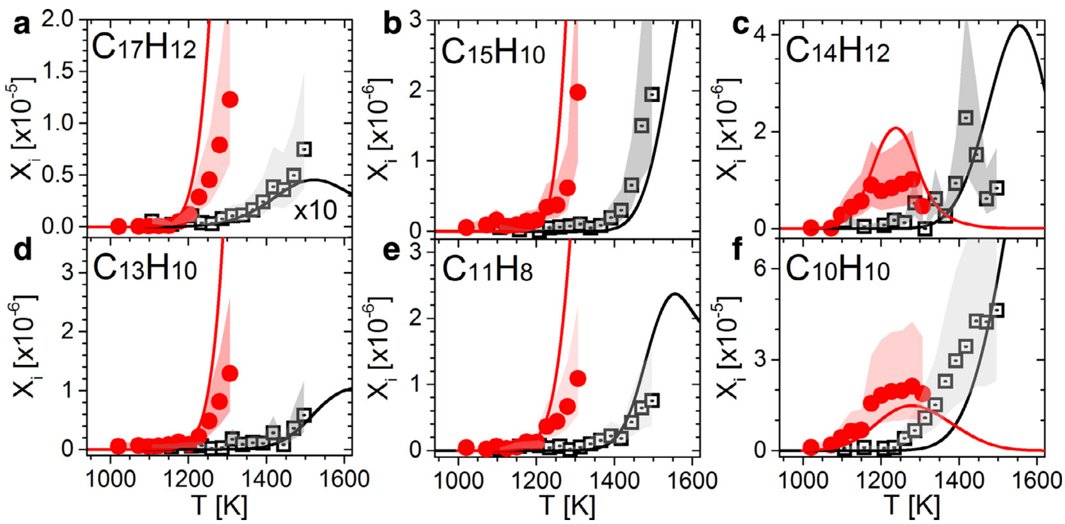


Fig. 4. Mole fraction profiles of uncommon PAHs measured in indene pyrolysis in a flow reactor at 30 (*black squares*) and 760 (*red circles*) Torr. Shadows with the data plots represent the uncertainty of the measurements. Lines indicate the model predictions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

sures. ROP analyses were performed under two different conditions at an indene conversion of 17.5%. The reaction networks in the decomposition of indene at both pressures are plotted in Fig. 5. The dominant product of indene decomposition is indenyl radical which accounts for more than 70% of indene consumption. The low C–H bond energy favors a competitive selectivity of H-elimination reaction compared to C–C bond dissociation reaction in unimolecular consumption channels. At higher pyrolysis pressure, molecules collide more frequently, so bimolecular H-abstraction reaction takes dominance in indene consumption.

Indanyl and 5-vinylidene-1,3-cyclopentadiene (C_7H_6) are minor indene consumption products. Rate of production (ROP) analysis in Fig. 5 shows that two indanyl radicals account for ~20% of the total carbon flux from indene, where the contribution of each channel does not change much as pressure increases. Unimolecular decomposition yielding C_7H_6 contributes less than 1% to the total indene consumption.

Sensitivity analysis for indene consumption is shown in Fig. 6. The mole fraction of indene is sensitive to the high-pressure limit of R1 at both pressures. The low-pressure limit of R1 is more sensitive at 30 Torr, while R2 is more sensitive at 760 Torr. H-addition reaction (R3) plays a negative effect on indene decomposition at higher pressure since it has a competition with R2 for H-atom consumption.

5.1.2. Decomposition of C_9 intermediates

ROP analysis (performed at 17.5% indene conversion) illustrates that indenyl decomposition is negligible, though three unimolecular dissociation pathways are included in the present model. Indenyl combination channels consumes 71% and 98% at 30 and 760 Torr, respectively. It should also be noted that the rate constants of indenyl dissociation reactions are estimated in literature models [39,56,57]. However, these reactions may impact PAH formation through the carbon flux competition of indenyl. Therefore, accurate quantum calculations of indenyl decomposition are required to adequately illustrate the kinetics of indenyl radical.

As shown in Fig. 5, 2-indanyl ($C_9H_9{}^2$), isomerized from 1-indanyl ($C_9H_9{}^1$), later opens its five-member ring to form phenyl-allyl ($C_6H_5C_3H_4$). At 30 Torr, indanyl mainly isomerizes to phenyl-allyl. When reaction pressure raises, more indanyl reacts

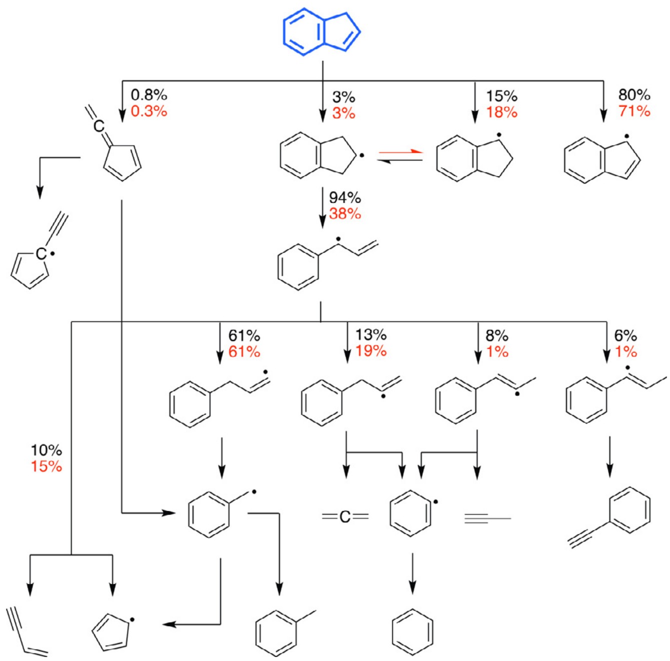


Fig. 5. Reaction flux analysis of indene decomposition at 17.5% conversion. Arrows indicate reaction steps; the numbers indicate the conversion ratios of each step at 30 (black) and 760 torr (red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

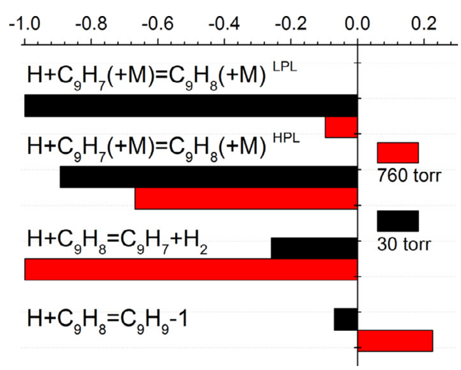


Fig. 6. Normalized sensitivity coefficient of indene consumption, performed at an indene conversion of 17.5% at 30 and 760 Torr. Only the most sensitive four reactions are listed. HPL and LPL are high- and low-pressure limits of R1. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

with indenyl and goes back to indene. Further isomerization and decomposition of phenyl-allyl contributes to benzyl (~60%), phenyl (~20%), cyclopentadienyl (<15%) and phenyl-acetylene (<6%). They are precursors of toluene, cyclopentadiene and benzene. Model predictions of these species are presented in Figs. 1 and 2.

5.1.3. Formation of small intermediates

Small intermediates, particularly C_2–C_5 hydrocarbons, are formed through various pathways *via* multi-elementary reaction steps. 5-Vinylidene-1,3-cyclopentadiene formed *via* the unimolecular decomposition of indene may lead to C_7H_7 or C_7H_5 radicals. C_7H_5 radical is mostly formed by the decomposition of C_7H_6; it further decomposes to C_4H_2, C_3H_3 and C_2H_2 [107] or contributes in PAH formation via combination reactions. Isomerization and dissociation of phenyl-allyl impact the formation of small intermediates, such as propargyl, allene, propyne, acetylene and vinyl-acetylene. Propargyl combines with cyclopentadienyl to form styrene which is a major formation pathway of styrene at 760 Torr. At 30 Torr,

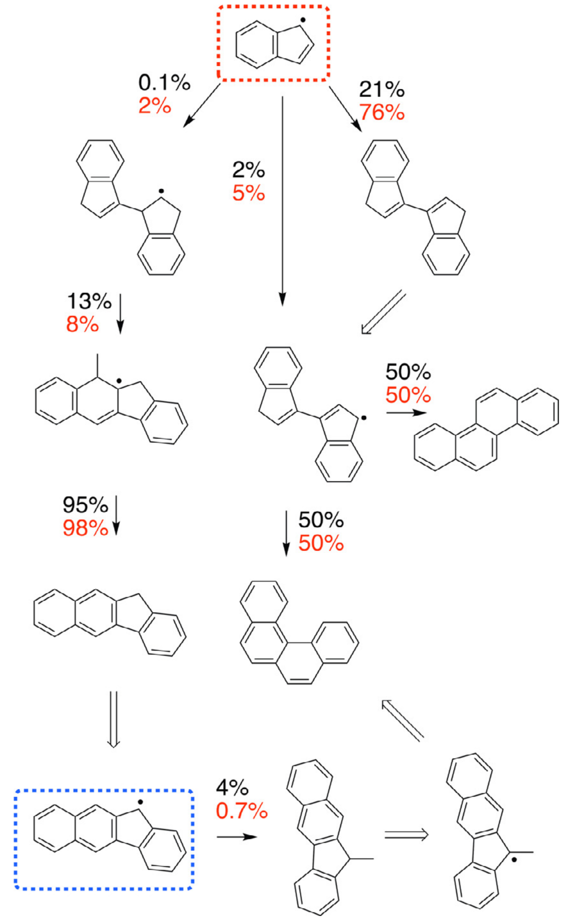


Fig. 7. ROP analysis performed for C_{17} and C_{18} species at 70% indene conversion. The blue frame highlights larger RSR generated by indenyl radical. Numbers beside the arrows are the contributions of this reaction step in the consumption of the precursors. Black and red colors separate 30 and 760 Torr conditions, respectively. Double-lined arrows indicate those reaction steps that consume >99.5% of the precursors. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

styrene is mainly formed *via* a complicated H-addition pathway from indanyl where phenyl-allyl and phenyl-propyl radicals are key intermediates in this process.

Model predictions of these small species are shown in Fig. 2. Only the data at 30 Torr are presented for active species, such as C_7H_6, C_7H_5, C_5H_5 and C_3H_3, since the concentration of these species is too low to be detected at 760 Torr. The mole fractions of small intermediates, such as allene, propyne, vinyl-acetylene and butadiyne, decrease as the pressure increases. It is because the formation of these species is more or less controlled by the isomerization of indanyl to phenyl-allyl, as shown in Fig. 5

5.1.4. Formation of common PAHs

Although the dominant contributor to PAH growth would be indenyl radical (Sections 5.1.4 and 5.1.5), we still want to know if other cyclopentadienyl-like or benzyl-like radicals have any contribution in this process (See Section 5.1.6). One point that needs to be clarified for the experiments of this work is that the mole fractions of PAHs are not calculated for every isomer due to their close ionization energies. As introduced in the experimental method Section 2 [68,80], additional measurements need to be performed at photon energies between the ionization energies of every two neighboring isomers. Due to the close ionization energies between isomers, the photon energy applied in the measurement would be very close to the ionization energy of the target isomer. In this

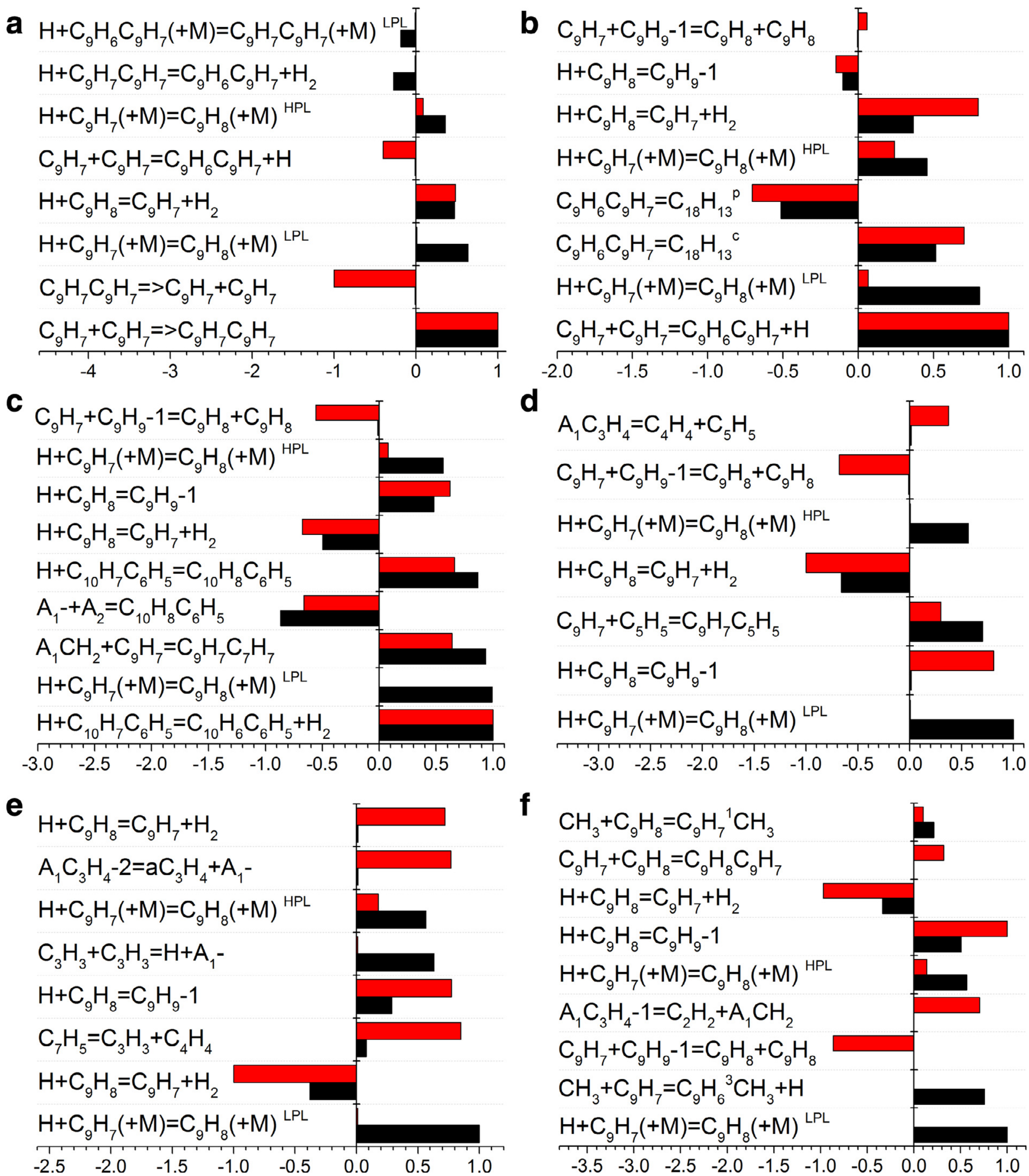


Fig. 8. Sensitivity analysis performed at 70% indene conversion on the selected PAH species: (a) 3,3-bi-indene, (b) chrysene, (c) fluoranthene, (d) phenanthrene, (e) acenaphthylene and (f) naphthalene. Black and red bars indicate 30 and 760 Torr conditions, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

case, the cross-section of this target isomer at the applied photon energy is very small, so that its signal would be easily influenced by the uncertainties of its isomers with larger cross sections. In general, this treatment may introduce large random uncertainty in the mole fraction values through measurement and mathematical process. Therefore, we instead decided to use one summed mole fraction of all PAH isomers with the same chemical formula when comparing with modeling results.

PAH species commonly discussed in previous PAH studies are presented in Fig. 3; these are $C_{18}H_{14}$, $C_{18}H_{12}$, $C_{16}H_{10}$, $C_{14}H_{10}$, $C_{12}H_8$ and $C_{10}H_8$. Figure 3(a) shows the model predictions of $C_{18}H_{14}$ where the measured mole fractions are calculated with the PICS of 3,3-bi-indene. The combination of two indenyl radicals is one of the most favorable reactions for indenyl. It consumes 21% and 76% of indenyl at 30 and 760 Torr, respectively, as shown in Fig. 7 ROP analysis at 70% indene conversion. It forms bi-indene isomers [63] which are represented by 3,3-bi-indene in the present model. 5- and 11-methyl-benzofluorene are minor $C_{18}H_{14}$ isomers. Further reactions of $C_{18}H_{14}$ yield $C_{18}H_{12}$, which can be chrysene, benzo-phenanthrene and triphenylene [10,61,63,65,108]. Model prediction shows that the quantities of chrysene and benzo-phenanthrene are comparable while that of triphenylene is smaller (chrysene

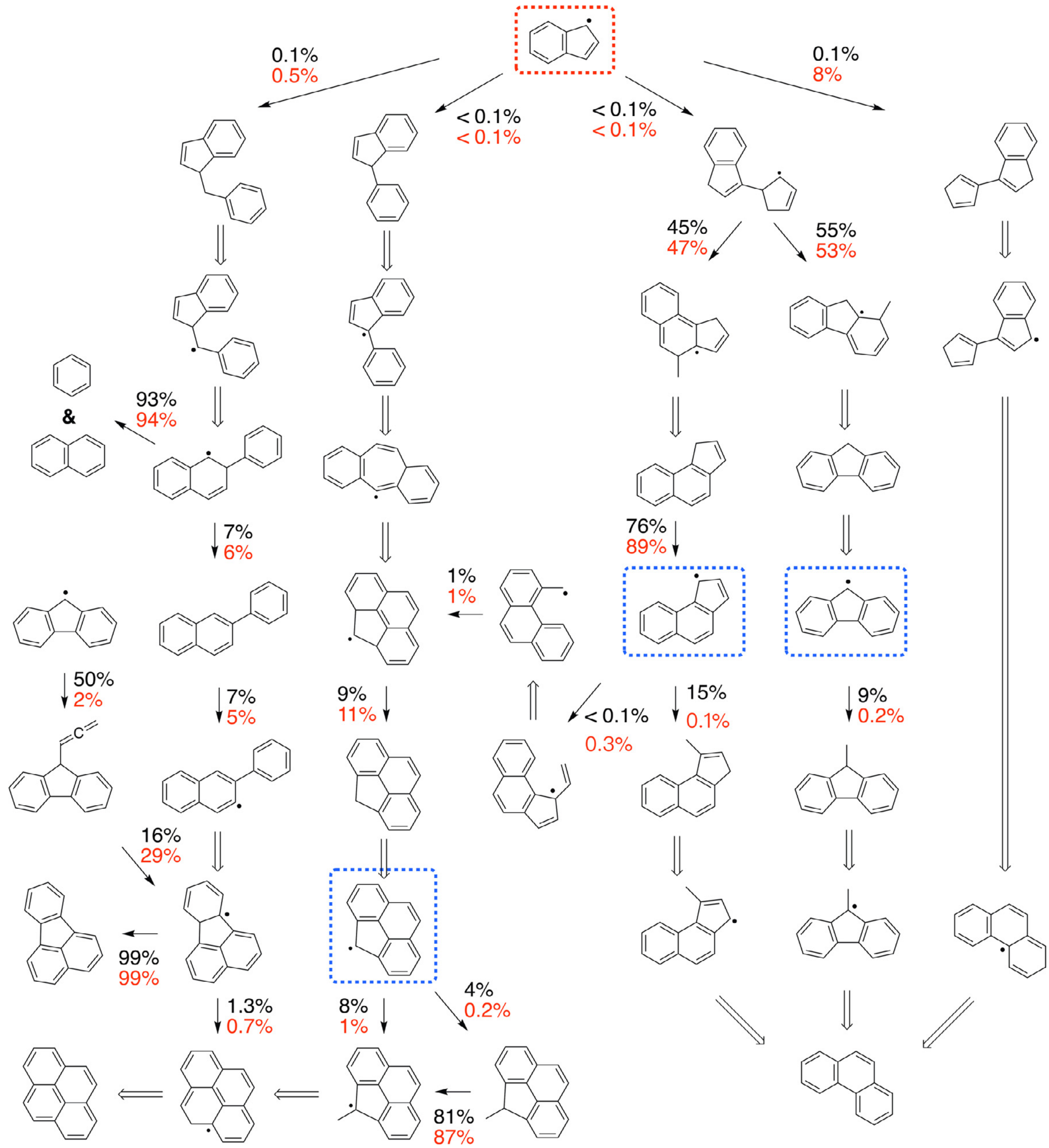


Fig. 9. ROP analysis performed for C_{13}–C_{16} species at 70% indene conversion. The blue frames highlight the larger RSRs generated by indenyl radical. Numbers beside the arrows are the contributions of this reaction step in the consumption of the precursors. Black and red colors separate 30 and 760 Torr conditions, respectively. Double-lined arrows indicate those reaction steps that consume >99.5% of the precursors. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

62%, benzo-phenanthrene 33%, and triphenylene 5% at ~1300 K, 760 Torr). Sensitivity analysis is performed at the temperature of 70% indene conversion at both pressures, as shown in Fig. 8. Besides the formation reactions (R88 for 3,3-bi-indene and R95 for chrysene), both species are sensitive to R99. It may be because R88 and R99 are two main branching pathways competing for the consumption of indenyl.

For $C_{16}H_{10}$ species, ROP analysis in Fig. 9 shows that indenyl reactions are the main formation source for fluoranthene and pyrene in all cases. The contributions from the reaction of benz-indenyl + propargyl to pyrene and fluorenyl + propargyl to fluoranthene are much smaller, and the HACA route (phenanthrenyl + acetylene) is negligible. Sensitivity analysis of fluoranthene in Fig. 8(c) shows that not only the combination of indenyl and benzyl (R63), but also the H-abstraction of phenyl-naphthalene (R69) and β-scission of C–H bond of $C_{10}H_8C_6H_5$ (R67) are rate controlling reactions of $C_{16}H_{10}$ species. R67 is a branching channel of this pathway. It competes with the C–C bond fission of $C_{10}H_8C_6H_5$ for naphthalene and phenyl. ROP analysis shows that R67 has a conversion ratio only around 7%.

$C_{14}H_{10}$ is calibrated as phenanthrene in the experiments and it has similar mole fraction to $C_{16}H_{10}$, shown in Fig. 3. Its dominant formation pathway is the bimolecular reaction of indenyl and cyclopentadienyl (R46), which consumes 8% of indenyl, as shown in Fig. 9. Sensitivity analysis in Fig. 8(d) also shows that R46 is the rate controlling step of phenanthrene formation pathways. The combination of phenyl + phenylacetylene and the decomposition of $C_{13}H_9CH_2$ radicals (benzindenyl-methyl or fluorenyl-methyl) have

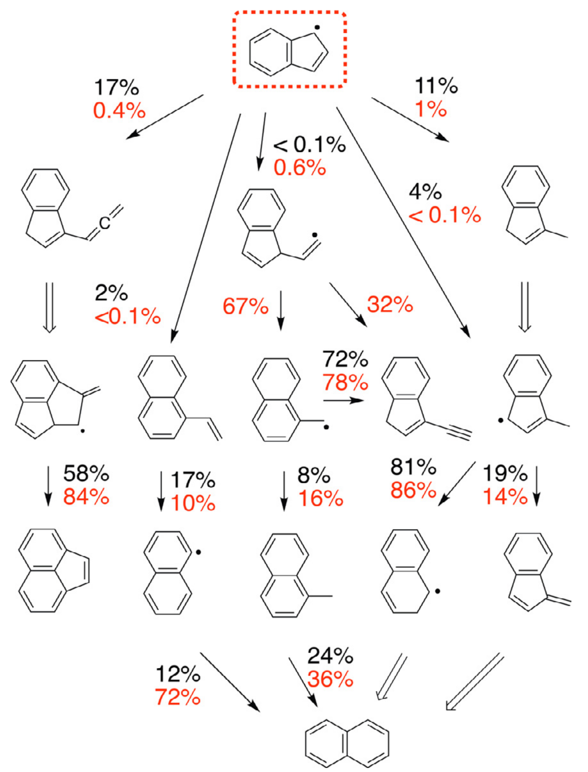


Fig. 10. ROP analysis performed for C_{10}–C_{12} species at 70% indene conversion. Numbers beside the arrows are the contributions of this reaction step in the consumption of the precursors. Black and red colors separate 30 and 760 Torr conditions, respectively. Double-lined arrows indicate those reaction steps that consume > 99.5% of the precursors. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

minor contributions. Another sensitive reaction is the decomposition of phenyl-allyl radical ($A_1C_3H_4$), which is a source of cyclopentadienyl radical.

Two $C_{12}H_8$ species are proposed in this study, i.e., acenaphthylene and ethynyl-naphthalene [40,109,110]. Both of these may be formed by the addition of acetylene on naphthyl radical, which are competing HACA channels. When comparing their contributions to indenyl routes, HACA routes are minor. The reaction of indenyl and propargyl (R27) forms more $C_{12}H_8$ species (> 80%). Branching channels of R27 and R35 lead to the formation of acenaphthylene and vinyl-naphthalene, whose ratios are shown in Fig. 10. Sensitivity analysis of acenaphthylene in Fig. 8(e) illustrates that the formation efficiency of propargyl significantly influences its concentration.

$C_{10}H_8$ is the summation of naphthalene, azulene and methylene-indene, which is calibrated as naphthalene in this experiments. $C_{10}H_8$ has higher mole fraction than $C_{16}H_{10}$, $C_{14}H_{10}$ and $C_{12}H_8$ but less than $C_{18}H_{12}$, as shown in Fig. 3. ROP analyses in Figs. 9 and 10 indicate that the main naphthalene formation pathway changes from methyl addition on indenyl to the decomposition of $C_{10}H_8C_6H_5$ radical. This radical is the product of indenyl and benzyl combination via R63-R66. The former pathway consumes 11% of indenyl at 30 Torr, and drops to 1% at 760 Torr. In contrast, the contribution of R63 increases 5 times with pressure. Sensitivity analysis of naphthalene also reflects this observation. R16 is very sensitive at 30 Torr, while benzyl formation reaction is very sensitive at 760 Torr.

In a general view of the sensitivity analysis of PAH species, shown in Fig. 8, we observe that the low-pressure limit (LPL) of R1 is always sensitive at low pressure, and R2 is sensitive at high pressure. On one hand, indenyl is mainly formed via R1 at low pressure, while on the other hand, R1 provides H atoms that enhance the formation of indanyl. It leads to small intermediates which are the source for addition reactions to PAHs. R2 has positive sensitivity for 3,3-bi-indene and chrysene, but negative for fluoranthene, phenanthrene, acenaphthylene and naphthalene. This is because H-abstraction of indene (R2) is competing with H-addition of indene (R3). The formation of latter four PAHs not only needs indenyl but also corresponding small intermediates.

5.1.5. Uncommon PAHs formation

Figure 4 shows some PAH species that are not commonly included in previous modeling studies. They are normally odd C-number species that are ignored in traditional HACA mechanism but are considered to be critical in this study *via* radical combination PAH pathways. $C_{17}H_{12}$ is is speculated to be benzo-fluorene [65], and its mole fraction increases significantly with the pressure. $m/z = 190$ is $C_{15}H_{10}$, and its molecular structure is proposed to be 4*H*-cyclopenta[def]phenanthrene in literature [111,112]. ROP analysis shows that its formation consumes less than 0.1% of indenyl. Since bi-indene is detected in this study, indenyl-cyclopentadiene, having similar reactivity, is also detectable by the same technique. It could be a candidate of $C_{14}H_{12}$ species. Other possible $C_{14}H_{12}$ isomers are methyl-fluorene and methyl-benzindene. Quantitative modeling results show that the mole fractions of the latter two isomers are higher than that of indenyl-cyclopentadiene. The reactions of $C_9H_8 + C_5H_5$ and $C_9H_7 + C_5H_6$ are not favored to produce indenyl-cyclopentadiene, but are more likely to form fluorene and benz-indene ($C_{13}H_{10}$), as shown by the ROP analysis in Fig. 9. We also observe from ROP analysis that $C_9H_7 + iC_4H_3$ reaction has considerable contribution in fluorene formation. $C_{11}H_8$ is calibrated as ethynyl-indene in the experiments [53,96,113]. In general, the proposed formation pathways in the present model reproduce the measured mole fraction quite well at both pressures, as shown in Fig. 4.

$C_{10}H_{10}$ is speculated as methyl-indene in this work [53,96,113]. The model under-predicts the formation of $C_{10}H_{10}$ in Fig. 4(f). It could be because of the unknown methyl-indene formation pathways or the under-prediction of methyl radical at temperatures around 1100 K. An important source of methyl radical is the reaction of indenyl and indene. It produces benzo-fluorene while releasing a methyl radicals. However, this part of the present model is analogized to the kinetics of cyclopentadienyl radical, which may introduce large uncertainty to methyl prediction. Further detailed calculations of this scheme at the potential energy surface of $C_{18}H_{15}$ may reveal the kinetics of benzo-fluorene and methyl-indene.

5.1.6. PAHs grown from other RSR radicals

Generally, PAH species can be divided into different aromatic groups: monocyclic, bicyclic, tricyclic, etc. Five-member-ring RSRs can be formed in every aromatic group as highlighted by the blue rectangles in Figs. 7 and 9. The smallest five-member-ring RSR is cyclopentadienyl radical. Indenyl radical studied in this work is a bicyclic five-member-ring RSR. With the addition of other five-member-ring species, the five-member-ring RSRs could grow bigger through self-incentive reaction processes. For example, indene is yielded from cyclopentadienyl radical [49,86,114] while fluorene and benz-indene are yielded *via* $C_9H_7 + C_5H_6$ or $C_5H_5 + C_9H_8$ [50]. Therefore, similar reactions are also possible for fluorenyl, benz-indenyl, benzo-fluorenyl and even larger five-member-ring RSRs. This efficient PAH growth moiety agrees with the radical chain reaction, recommended recently by Johansson et al. [59].

Besides the five-member-ring RSRs, the contribution of other RSRs is not so remarkable in indene pyrolysis, which can be ascribed to the fuel selectivity of PAH growth pathways. Propargyl, benzyl, tropyl, vinyl-cyclopentadienyl, and other large RSRs

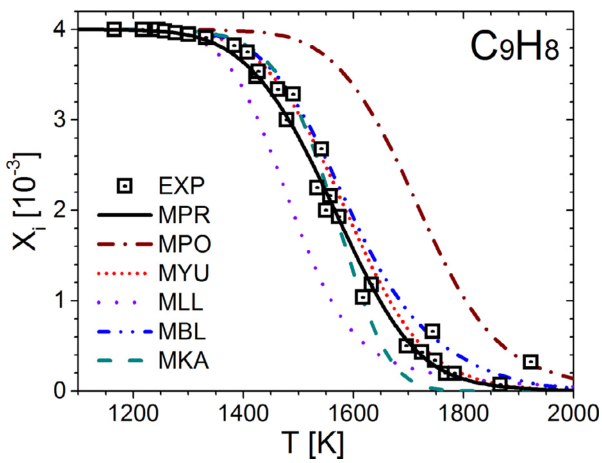


Fig. 11. Modeling results of indene decomposition in a shock tube; experimental data taken from [62]. Simulated results were shifted by 80 K according to the literature [120,121]. MPR: present model; MPO: Polimi model [81]; MYU: Yuan et al.'s model [9]; MLL: Lawrence Livermore model [58]; MBL: Blanquart et al.'s model [7]; MKA: Wang et al.'s model [8].

with similar structures are potential contributors. For example, in Figs. 9 and 10, the addition reactions of propargyl on indenyl and fluorenyl are quite efficient to build larger PAHs. Detailed chemical kinetics for some of these were investigated in previous studies [31,42,45,47,54,85,115–119]; however, there is a lack of knowledge about more other RSRs, especially large ones, which may be comparatively as important as indenyl in PAH growth.

5.2. Indene pyrolysis in shock tube

Laskin and Liftshitz studied indene pyrolysis in a shock tube at a pressure of 3.7 atm over a temperature range of 1150–1900 K [62]. Quantitative species measurements were provided by gas chromatography (GC). We adopted this dataset to validate the present model. Comparisons between experimental data and simulations are presented in Figs. 11 and 12. Note that simulated temperatures were shifted by 80 K to match indene conversion [120,121] in order to compare experimental and simulated branching fractions for an equivalent conversion. The present model adequately captures the mole fractions of most species; only a few species, like naphthalene, methylene-indene and benzene, are slightly over-predicted. Predictions by literature models are also compared in Figs. 11 and 12. Generally, literature models cannot capture the indene decomposition and the intermediates concentrations in the same time.

6. Conclusions

Pyrolysis of indene was investigated in a flow reactor at 30 and 760 Torr over the temperature range of 975–1450 K using SVUV-PIMS. A number of pyrolytic species were quantified, such as free radicals, various small hydrocarbons and PAHs. Their mole fraction profiles were measured as a function of reactor temperature. Based on these experimental observations, an improved kinetic model was developed based on previous PAH models [7,9,10,13–15,56,57,66,77]. We adopted recent theoretical progress and added some necessary reaction schemes in our model. The present model was then validated against the flow reactor data from this study and datasets reported in literature [62]. By performing ROP and sensitivity analysis, critical reaction pathways were identified for indene pyrolysis, particularly those leading to PAH formation from indenyl radical.

Under the pyrolysis condition, indene is mainly consumed by unimolecular C–H bond dissociation and H-abstraction reactions. Both of these pathways have comparable contribution from low to atmospheric pressure. Subsequent reactions of indenyl favor the PAH channels, since indenyl is not easy to decompose at the conditions of this study. The dominant PAH formation pathways in this study are the bi-molecular addition reactions of indenyl

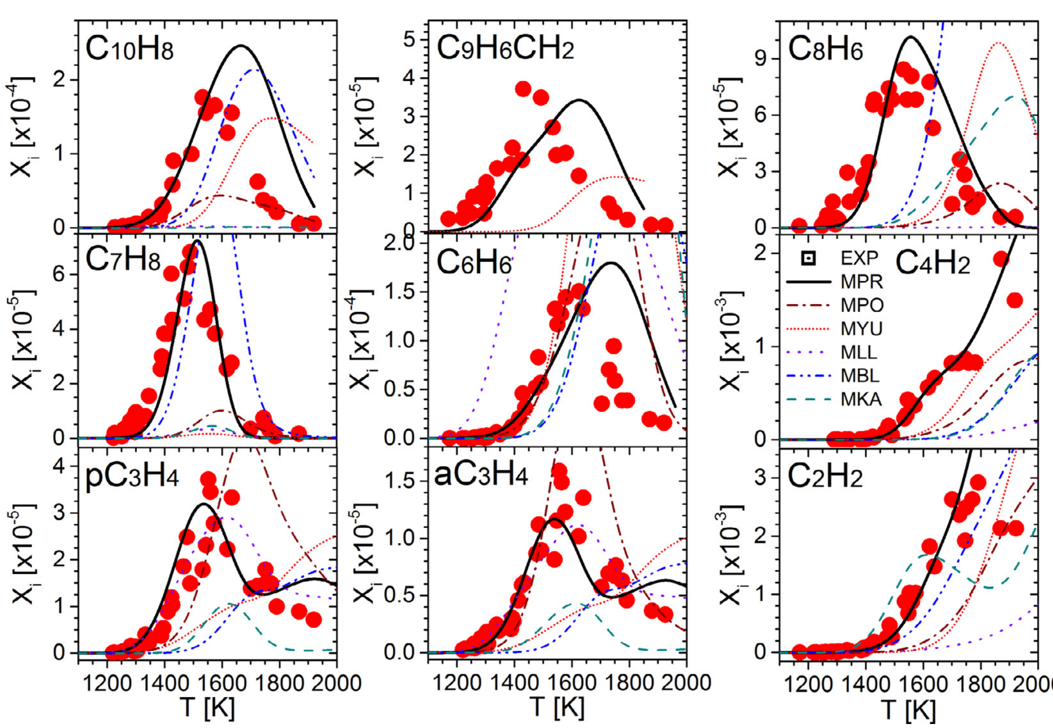


Fig. 12. Predicted mole fraction profiles of the species measured in shock tube indene pyrolysis [62]. Simulated results were shifted by 80 K according to the literature [120,121]. MPR: present model; MPO: Polimi model [81]; MYU: Yuan et al.'s model [9]; MLL: Lawrence Livermore model [58]; MBL: Blanquart et al.'s model [7]; MKA: Wang et al.'s model [8].

radical with indene or a series of other intermediates. PAH growth also has particular contribution from the reactions of other resonantly stabilized radicals with five-member ring in their molecular structures, such as cyclopentadienyl, fluorenyl, benz-indenyl and benzo-fluorenyl. Radical chain reactions provide huge passage for PAH growth form one RSR to larger ones. Typically, the cyclical addition of acetylene on indenyl yields vinyl-indenyl or benz-indenyl, while the cyclical addition of indene on indenyl yields benzo-fluorenyl and even larger RSRs, although these RSRs are not detected in this study. Further PAH modeling studies should refine the reaction network of other typical RSRs besides indenyl radical, which may help provide prediction of not only the concentration but also the chemical and/or physical structures of PAHs and soot.

Acknowledgment

Research reported in this publication was funded by the Office of Sponsored Research at King Abdullah University of Science and Technology (KAUST), Key Science Foundation of Higher Education of Henan (19A480002) and National Natural Science Foundation of China (91541201 and 51622605). In addition, Hanfeng would also like to thank Xiaoyuan, in particular, for the encouragement, patience and love from her over the passed years.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2019.04.040.

References

[1] A. Ebenstein, M. Fan, M. Greenstone, G. He, M. Zhou, New evidence on the impact of sustained exposure to air pollution on life expectancy from China's Huai River Policy, Proc. Natl. Acad. Sci. U. S. A. 114 (2017) 10384–10389.

[2] I.C. Pope, R.T. Burnett, M.J. Thun, et al., Lung cancer, cardiopulmonary mortality, and long-term exposure to fine particulate air pollution, JAMA 287 (2002) 1132–1141.

[3] H. Richter, J.B. Howard, Formation of polycyclic aromatic hydrocarbons and their growth to soot – a review of chemical reaction pathways, Prog. Energy Combust. Sci. 26 (2000) 565–608.

[4] C.S. McEnally, L.D. Pfefferle, B. Atakan, K. Kohse-Hoinghaus, Studies of aromatic hydrocarbon formation mechanisms in flames: progress towards closing the fuel gap, Prog. Energy Combust. Sci. 32 (2006) 247–294.

[5] H. Wang, Formation of nascent soot and other condensed-phase materials in flames, Proc. Combust. Inst. 33 (2011) 41–67.

[6] K. Kohse-Höinghaus, Clean combustion: chemistry and diagnostics for a systems approach in transportation and energy conversion, Prog. Energy Combust. Sci. 65 (2018) 1–5.

[7] G. Blanquart, P. Pepiot-Desjardins, H. Pitsch, Chemical mechanism for high temperature combustion of engine relevant fuels with emphasis on soot precursors, Combust. Flame 156 (2009) 588–607.

[8] Y. Wang, A. Raj, S.H. Chung, A PAH growth mechanism and synergistic effect on PAH formation in counterflow diffusion flames, Combust. Flame 160 (2013) 1667–1676.

[9] W.H. Yuan, Y.Y. Li, P. Dagaut, J.Z. Yang, F. Qi, Investigation on the pyrolysis and oxidation of toluene over a wide range conditions. II. A comprehensive kinetic modeling study, Combust. Flame 162 (2015) 22–40.

[10] N.A. Slavinskaya, P. Frank, A modelling study of aromatic soot precursors formation in laminar methane and ethene flames, Combust. Flame 156 (2009) 1705–1722.

[11] H. Richter, W.J. Grieco, J.B. Howard, Formation mechanism of polycyclic aromatic hydrocarbons and fullerenes in premixed benzene flames, Combust. Flame 119 (1999) 1–22.

[12] N.M. Marinov, W.J. Pitz, C.K. Westbrook, M.J. Castaldi, S.M. Senkan, Modeling of aromatic and polycyclic aromatic hydrocarbon formation in premixed methane and ethane flames, Combust. Sci. Technol. 116-117 (1996) 211–287.

[13] J. Appel, H. Bockhorn, M. Frenklach, Kinetic modeling of soot formation with detailed chemistry and physics: laminar premixed flames of C2 hydrocarbons, Combust. Flame 121 (2000) 122–136.

[14] H. Wang, M. Frenklach, A detailed kinetic modeling study of aromatics formation in laminar premixed acetylene and ethylene flames, Combust. Flame 110 (1997) 173–221.

[15] H. Jin, A. Frassoldati, Y. Wang, X. Zhang, M. Zeng, Y. Li, F. Qi, A. Cuoci, T. Faravelli, Kinetic modeling study of benzene and PAH formation in laminar methane flames, Combust. Flame 162 (2015) 1692–1711.

[16] H. Wang, M. Frenklach, Calculations of rate coefficients for the chemically activated reactions of acetylene with vinylic and aromatic radicals, J. Phys. Chem. 98 (1994) 11465–11489.

[17] F. Xu, K.C. Lin, G.M. Faeth, Soot formation in laminar premixed methane/oxygen flames at atmospheric pressure, Combust. Flame 115 (1998) 195–209.

[18] N.E. Sánchez, A. Callejas, Á. Millera, R. Bilbao, M.U. Alzueta, Polycyclic Aromatic hydrocarbon (PAH) and soot formation in the pyrolysis of acetylene and ethylene: effect of the reaction temperature, Energy Fuels 26 (2012) 4823–4829.

[19] C. Saggese, N.E. Sánchez, A. Frassoldati, A. Cuoci, T. Faravelli, M.U. Alzueta, E. Ranzi, Kinetic modeling study of polycyclic aromatic hydrocarbons and soot formation in acetylene pyrolysis, Energy Fuels 28 (2013) 1489–1501.

[20] S.A. Skeen, B. Yang, H.A. Michelsen, J.A. Miller, A. Violi, N. Hansen, Studies of laminar opposed-flow diffusion flames of acetylene at low-pressures with photoionization mass spectrometry, Proc. Combust. Inst. 34 (2013) 1067–1075.

[21] J.D. Bittner, J.B. Howard, Composition profiles and reaction mechanisms in a near-sooting premixed benzene/oxygen/argon flame, Proc. Combust. Inst. 18 (1981) 1105–1116.

[22] J. Griesheimer, K.H. Homann, Large molecules, radicals ions, and small soot particles in fuel-rich hydrocarbon flames: part II. Aromatic radicals and intermediate PAHs in a premixed low-pressure naphthalene/oxygen/argon flame, Proc. Combust. Inst. 27 (1998) 1753–1759.

[23] R.P. Lindstedt, G. Skevis, A.R. Burgess, F.L. Dryer, Benzene formation chemistry in premixed 1,3-butadiene flames, Proc. Combust. Inst. 26 (1996) 703–709.

[24] J.A. Miller, C.F. Melius, Kinetic and thermodynamic issues in the formation of aromatic compounds in flames of aliphatic fuels, Combust. Flame 91 (1992) 21–39.

[25] E.W. Diau, M.C. Lin, C.F. Melius, A theoretical study of the $CH_3+C_2H_2$ reaction, J. Chem. Phys. 101 (1994) 3923–3927.

[26] A.M. Dean, Detailed kinetic modeling of autocatalysis in methane pyrolysis, J. Phys. Chem. 94 (1990) 1432–1439.

[27] M.B. Colket, D.J. Seery, Reaction mechanisms for toluene pyrolysis, Proc. Combust. Inst. 25 (1994) 883–891.

[28] C.F. Melius, M.E. Colvin, N.M. Marinov, W.J. Pitz, S.M. Senkan, Reaction mechanisms in aromatic hydrocarbon formation involving the C5H5 cyclopentadienyl moiety, Proc. Combust. Inst. 26 (1996) 685–692.

[29] N.M. Marinov, W.J. Pitz, C.K. Westbrook, A.M. Vincitore, M.J. Castaldi, S.M. Senkan, C.F. Melius, Aromatic and polycyclic aromatic hydrocarbon formation in a laminar premixed n-butane flame, Combust. Flame 114 (1998) 192–213.

[30] A. D'Anna, J.H. Kent, Aromatic formation pathways in non-premixed methane flames, Combust. Flame 132 (2003) 715–722.

[31] C.L. Rasmussen, M.S. Skjøth-Rasmussen, A.D. Jensen, P. Glarborg, Propargyl recombination: estimation of the high temperature, low pressure rate constant from flame measurements, Proc. Combust. Inst. 30 (2005) 1023–1031.

[32] N. Hansen, J.A. Miller, P.R. Westmoreland, T. Kasper, K. Kohse-Höinghaus, J. Wang, T.A. Cool, Isomer-specific combustion chemistry in allene and propyne flames, Combust. Flame 156 (2009) 2153–2164.

[33] M.R. Djokic, K.M. Van Geem, C. Cavallotti, A. Frassoldati, E. Ranzi, G.B. Marin, An experimental and kinetic modeling study of cyclopentadiene pyrolysis: first growth of polycyclic aromatic hydrocarbons, Combust. Flame 161 (2014) 2739–2751.

[34] R.K. Robinson, R.P. Lindstedt, On the chemical kinetics of cyclopentadiene oxidation, Combust. Flame 158 (2011) 666–686.

[35] R.G. Butler, I. Glassman, Cyclopentadiene combustion in a plug flow reactor near 1150 K, Proc. Combust. Inst. 32 (2009) 395–402.

[36] A. Lifshitz, C. Tamburu, A. Suslensky, F. Dubnikova, Decomposition and ring expansion in methylcyclopentadiene: single-pulse shock tube and modeling study, Proc. Combust. Inst. 30 (2005) 1039–1047.

[37] J. Filley, J.T. McKinnon, Dimerization of cyclopentadienyl radical to produce naphthalene, Combust. Flame 124 (2001) 721–723.

[38] R.P. Lindstedt, K.A. Rizos, The formation and oxidation of aromatics in cyclopentene and methyl-cyclopentadiene mixtures, Proc. Combust. Inst. 29 (2002) 2291–2298.

[39] N.A. Slavinskaya, U. Riedel, S.B. Dworkin, M.J. Thomson, Detailed numerical modeling of PAH formation and growth in non-premixed ethylene and ethane flames, Combust. Flame 159 (2012) 979–995.

[40] B. Shukla, M. Koshi, Comparative study on the growth mechanisms of PAHs, Combust. Flame 158 (2011) 369–375.

[41] N. Hansen, M. Schenk, K. Moshammer, K. Kohse-Höinghaus, Investigating repetitive reaction pathways for the formation of polycyclic aromatic hydrocarbons in combustion processes, Combust. Flame 180 (2017) 250–261.

[42] J.A. Miller, S.J. Klippenstein, The recombination of propargyl radicals and other reactions on a C_6H_6 potential, J. Phys. Chem. A 107 (2003) 7783–7799.

[43] V.V. Kislov, A.M. Mebel, The formation of naphthalene, azulene, and fulvalene from cyclic C-5 species in combustion: an ab Initio/RRKM study of 9-h-fulvalenyl (C5H5-C5H4) radical rearrangements, J. Phys. Chem. A 111 (2007) 9532–9543.

[44] A.M. Mebel, V.V. Kislov, Can the C5H5 + C5H5 → C10H10 → C10H9 + H/C10H8 + H2 Reaction Produce Naphthalene? An Ab Initio/RRKM study, J. Phys. Chem. A 113 (2009) 9825–9833.

[45] A. Matsugi, A. Miyoshi, Computational study on the recombination reaction between benzyl and propargyl radicals, Int. J. Chem. Kinet. 44 (2012) 206–218.

[46] C. Cavallotti, D. Polino, On the kinetics of the C_5H_5+C_5H_5 reaction, Proc. Combust. Inst. 34 (2013) 557–564.
[47] S. Sinha, R.K. Rahman, A. Raj, On the role of resonantly stabilized radicals in polycyclic aromatic hydrocarbon (PAH) formation: pyrene and fluoranthene formation from benzyl-indenyl addition, Phys. Chem. Chem. Phys. 19 (2017) 19262–19278.
[48] A.E. Long, S.S. Merchant, A.G. Vandeputte, H.-H. Carstensen, A.J. Vervust, G.B. Marin, K.M. Van Geem, W.H. Green, Pressure dependent kinetic analysis of pathways to naphthalene from cyclopentadienyl recombination, Combust. Flame 187 (2018) 247–256.
[49] D. Wang, A. Violi, D.H. Kim, J.A. Mullholland, Formation of naphthalene, indene, and benzene from cyclopentadiene pyrolysis: a DFT study, J. Phys. Chem. A 110 (2006) 4719–4725.
[50] J.A. Mulholland, M. Lu, D.-H. Kim, Pyrolytic growth of polycyclic aromatic hydrocarbons by cyclopentadienyl moieties, Proc. Combust. Inst. 28 (2000) 2593–2599.
[51] S. Sharma, W.H. Green, Computed rate coefficients and product yields for c-C5H5 + CH3 → Products, J. Phys. Chem. A 113 (2009) 8871–8882.
[52] R.S. Tranter, S.J. Klippenstein, L.B. Harding, B.R. Giri, X.L. Yang, J.H. Kiefer, Experimental and theoretical investigation of the self-reaction of phenyl radicals, J. Phys. Chem. A 114 (2010) 8240–8261.
[53] A.M. Mebel, A. Landera, R.I. Kaiser, Formation mechanisms of naphthalene and indene: from the interstellar medium to combustion flames, J. Phys. Chem. A 121 (2017) 901–926.
[54] A. Raj, M.J. Al Rashidi, S.H. Chung, S.M. Sarathy, PAH growth initiated by propargyl addition: mechanism development and computational kinetics, J. Phys. Chem. A 118 (2014) 2865–2885.
[55] W. Yuan, Y. Li, P. Dagaut, J. Yang, F. Qi, Experimental and kinetic modeling study of styrene combustion, Combust. Flame 162 (2015) 1868–1883.
[56] W.H. Yuan, Y.Y. Li, P. Dagaut, J.Z. Yang, F. Qi, Investigation on the pyrolysis and oxidation of toluene over a wide range conditions. I. Flow reactor pyrolysis and jet stirred reactor oxidation, Combust. Flame 162 (2015) 3–21.
[57] A. Matsugi, A. Miyoshi, Modeling of two- and three-ring aromatics formation in the pyrolysis of toluene, Proc. Combust. Inst. 34 (2013) 269–277.
[58] H. Nakamura, D. Darcy, M. Mehl, C.J. Tobin, W.K. Metcalfe, W.J. Pitz, C.K. Westbrook, H.J. Curran, An experimental and modeling study of shock tube and rapid compression machine ignition of n-butylbenzene/air mixtures, Combust. Flame 161 (2014) 49–64.
[59] K.O. Johansson, M.P. Head-Gordon, P.E. Schrader, K.R. Wilson, H.A. Michelsen, Resonance-stabilized hydrocarbon-radical chain reactions may explain soot inception and growth, Science 361 (2018) 997–1000.
[60] H. Jin, G. Wang, Y. Wang, X. Zhang, Y. Li, Z. Zhou, J. Yang, F. Qi, Experimental and kinetic modeling study of laminar coflow diffusion methane flames doped with iso -butanol, Proc. Combust. Inst. 36 (2017) 1259–1267.
[61] A. Ajaz, A.C. Voukides, K.J. Cahill, R. Thamatam, S.L. Skraba-Joiner, R.P. Johnson, Microwave flash pyrolysis: C9H8 interconversions and dimerisations, Aust. J. Chem. 67 (2014) 1301–1308.
[62] A. Laskin, A. Lifshitz, Thermal decomposition of indene. Experimental results and kinetic modeling, Proc. Combust. Inst. 27 (1998) 313–320.
[63] C. Wentrup, H.-W. Winter, D. Kvaskoff, C9H8 Pyrolysis. o-Tolylacetylene, Indene, 1-Indenyl, and biindenyls and the mechanism of indene pyrolysis, J. Phys. Chem. A 119 (2015) 6370–6376.
[64] C. Cavallotti, S. Mancarella, R. Rota, S. Carrà, Conversion of C5 into C6 cyclic species through the formation of C7 intermediates†, J. Phys. Chem. A 111 (2007) 3959–3969.
[65] M. Lu, J.A. Mulholland, Aromatic hydrocarbon growth from indene, Chemosphere 42 (2001) 625–633.
[66] A. Lifshitz, C. Tamburu, A. Suslensky, F. Dubnikova, Decomposition, isomerization, and ring expansion in 2-methylindene: Single-pulse shock tube and modeling study, J. Phys. Chem. A 108 (2004) 3430–3438.
[67] G.M. Badger, R.W.L. Kimber, The formation of aromatic hydrocarbons at high temperatures part VII the pyrolysis of indene, J. Chem. Soc. (1960) 2746–2749.
[68] F. Qi, Combustion chemistry probed by synchrotron VUV photoionization mass spectrometry, Proc. Combust. Inst. 34 (2013) 33–63.
[69] Z. Zhou, X. Du, J. Yang, Y. Wang, C. Li, S. Wei, L. Du, Y. Li, F. Qi, Q. Wang, The vacuum ultraviolet beamline/endstations at NSRL dedicated to combustion research, J. Synchrot. Radiat. 23 (2016) 1035–1045.
[70] J. Cai, W. Yuan, L. Ye, Z. Cheng, Y. Wang, L. Zhang, F. Zhang, Y. Li, F. Qi, Experimental and kinetic modeling study of 2-butanol pyrolysis and combustion, Combust. Flame 160 (2013) 1939–1957.
[71] Y. Zhao, D.G. Truhlar, The M06 suite of density functionals for main group thermochemistry, thermochemical kinetics, noncovalent interactions, excited states, and transition elements: two new functionals and systematic testing of four M06-class functionals and 12 other functionals, Theor. Chem. Acc. 120 (2008) 215–241.
[72] B.J. Lynch, Y. Zhao, D.G. Truhlar, Effectiveness of diffuse basis functions for calculating relative energies by density functional theory, J. Phys. Chem. A 107 (2003) 1384–1388.
[73] R. Krishnan, J.S. Binkley, R. Seeger, J.A. Pople, Self-consistent molecular orbital methods. XX. A basis set for correlated wave functions, J. Chem. Phys. 72 (1980) 650–654.
[74] M.J. Frisch, G.W. Trucks, H.B. Schlegel, G.E. Scuseria, M.A. Robb, J.R. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G.A. Petersson, H. Nakatsuji, M. Caricato, X. Li, H.P Hratchian, A.F Izmaylov, J. Bloino, G. Zheng, J.L. Sonnenberg, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, J.A. Montgomery Jr., J.E. Peralta, F. Ogliaero, M. Bearpark, J.J. Heyd, E. Brothers, K.N. Kudin, V.N. Staroverov, R. Kobayashi, T. Keith, J. Normand, K. Raghavachari, A. Rendell, J.C Burant, S.S. Iyengar, J. Tomasi, M. Cossi, N Rega, J.M. Millam, M. Klene, J.E. Knox, J.B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R.E. Stratmann, O. Yazyev, A.J. Austin, R. Cammi, C. Pomelli, J.W. Ochterski, R.L. Martin, K. Morokuma, V.G. Zakrzewski, G.A. Voth, P. Salvador, J.J. Dannenberg, S. Dapprich, A.D. Daniels, O. Farkas, J.B. Foresman, J.V. Ortiz, J. Cioslowski, D.J. Fox, Gaussian 09, Gaussian, Inc., Wallingford, CT, 2009 revision C.01.
[75] Y. Zhao, R. Peverati, K. Tang, S. Luo, H.S. Yu, X. He, D.G. Truhlar, Minnesota Gaussian functional module MN-GFM 6.7, Department of Chemistry, University of Minnesota, Minneapolis, MN USA, 2015 https://comp.chem.umn.edu/MN-GFM/.
[76] L. Xing, J.L. Bao, Z. Wang, F. Zhang, D.G. Truhlar, Degradation of carbonyl hydroperoxides in the atmosphere and in combustion, J. Am. Chem. Soc. 139 (2017) 15821–15835.
[77] E. Pousse, Z.Y. Tian, P.A. Glaude, R. Fournet, F. Battin-Leclerc, A lean methane premixed laminar flame doped with components of diesel fuel part III: indane and comparison between n-butylbenzene, n-propylcyclohexane and indane, Combust. Flame 157 (2010) 1236–1260.
[78] F. Dubnikova, A. Lifshitz, Ring expansion and isomerization in methyl indene and methylene indene radicals. Quantum chemical and transition-state theory calculations, Isr. J. Chem. 43 (2003) 325–338.
[79] S. Fascella, C. Cavallotti, R. Rota, S. Carrà, The peculiar kinetics of the reaction between acetylene and the cyclopentadienyl radical, J. Phys. Chem. A 109 (2005) 7546–7557.
[80] Y.Y. Li, L.D. Zhang, Z.D. Wang, L.L. Ye, J.H. Cai, Z.J. Cheng, F. Qi, Experimental and kinetic modeling study of tetralin pyrolysis at low pressure, Proc. Combust. Inst. 34 (2013) 1739–1848.
[81] C. Saggese, A. Frassoldati, A. Cuoci, T. Faravelli, E. Ranzi, A wide range kinetic modeling study of pyrolysis and oxidation of benzene, Combust. Flame 160 (2013) 1168–1190.
[82] H. Wang, E. Dames, B. Sirjean, D.A. Sheen, R. Tango, A. Violi, J.Y.W. Lai, F.N. Egolfopoulos, D.F. Davidson, R.K. Hanson, C.T. Bowman, C.K. Law, W. Tsang, N.P. Cernansky, D.L. Miller, R.P. Lindstedt, A high-temperature chemical kinetic model of n-alkane (up to n-dodecane), cyclohexane, and methyl-, ethyl-, n-propyl and n-butyl-cyclohexane oxidation at high temperatures, JetSurF Version 2.0 (2010). September 19 http://web.stanford.edu/group/haiwanglab/JetSurF/JetSurF2.0/index.html .
[83] G. da Silva, J.A. Cole, J.W. Bozzelli, Kinetics of the cyclopentadienyl + acetylene, fulvenallene + H, and 1-ethynylcyclopentadiene + H reactions, J. Phys. Chem. A 114 (2010) 2275–2283.
[84] D. Polino, C. Cavallotti, Fulvenallene decomposition kinetics, J. Phys. Chem. A 115 (2011) 10281–10289.
[85] C. Cavallotti, M. Derudi, R. Rota, On the mechanism of decomposition of the benzyl radical, Proc. Combust. Inst. 32 (2009) 115–121.
[86] V.V. Kislov, A.M. Mebel, An ab initio G3-type/statistical theory study of the formation of indene in combustion flames. II. The pathways originating from reactions of cyclic C-5 species - Cyclopentadiene and cyclopentadienyl radicals, J. Phys. Chem. A 112 (2008) 700–716.
[87] K. Narayanaswamy, H. Pitsch, P. Pepiot, A chemical mechanism for low to high temperature oxidation of methylcyclohexane as a component of transportation fuel surrogates, Combust. Flame 162 (2015) 1193–1213.
[88] L. Vereecken, J. Peeters, Reactions of chemically activated C_9H_9 species II: the reaction of phenyl radicals with allene and cyclopropene, and of benzyl radicals with acetylene, Phys. Chem. Chem. Phys. 5 (2003) 2807–2817.
[89] H.A. Gueniche, P.A. Glaude, R. Fournet, F. Battin-Leclerc, Rich methane premixed laminar flames doped by light unsaturated hydrocarbons – III. Cyclopentene, Combust. Flame 152 (2008) 245–261.
[90] M.R. Harper, K.M. Van Geem, S.P. Pyl, G.B. Marin, W.H. Green, Comprehensive reaction mechanism for n-butanol pyrolysis and combustion, Combust. Flame 158 (2011) 16–41.
[91] K. Roy, M. Braun-Unkhoff, P. Frank, T. Just, Kinetics of the cyclopentadiene decay and the recombination of cyclopentadienyl radicals with H-atoms: enthalpy of formation of the cyclopentadienyl radical, Int. J. Chem. Kinet. 33 (2001) 821–833.
[92] H. Richter, J.B. Howard, Formation and consumption of single-ring aromatic hydrocarbons and their precursors in premixed acetylene, ethylene and benzene flames, Phys. Chem. Chem. Phys. 4 (2002) 2038–2055.
[93] B. Heyberger, N. Belmekki, V. Conraud, P.A. Glaude, R. Fournet, F. Battin-Leclerc, Oxidation of small alkenes at high temperature, Int. J. Chem. Kinet. 34 (2002) 666–677.
[94] B. Sirjean, P.A. Glaude, M.F. Ruiz-Lopez, R. Fournet, Theoretical kinetic study of thermal unimolecular decomposition of cyclic alkyl radicals, J. Phys. Chem. A 112 (2008) 11598–11610.
[95] A. Ratkiewicz, T.N. Truong, Kinetics of the C–C bond beta scission reactions in alkyl radical reaction class, J. Phys. Chem. A 116 (2012) 6643–6654.
[96] A.M. Mebel, Y. Georgievskii, A.W. Jasper, S.J. Klippenstein, Pressure-dependent rate constants for PAH growth: formation of indene and its conversion to naphthalene, Faraday Discuss. 195 (2016) 637–670.
[97] H. Richter, S. Granata, W.H. Green, J.B. Howard, Detailed modeling of PAH and soot formation in a laminar premixed benzene/oxygen/argon low-pressure flame, Proc. Combust. Inst. 30 (2005) 1397–1405.
[98] Y. Li, J. Cai, L. Zhang, T. Yuan, K. Zhang, F. Qi, Investigation on chemical structures of premixed toluene flames at low pressure, Proc. Combust. Inst. 33 (2011) 593–600.

[99] A. Cristadoro, S.U. Kulkarni, W.A. Burgess, E.G. Cervo, H.J. Räder, K. Müllen, D.A. Bruce, M.C. Thies, Structural characterization of the oligomeric constituents of petroleum pitches, Carbon 47 (2009) 2358–2370.
[100] R. Gredel, Y. Carpentier, G. Rouille, M. Steglich, F. Huisken, T. Henning, Abundances of PAHs in the ISM: confronting observations with experimental results, Astron. Astrophys. 530 (2011) 15.
[101] H. Wang, X. You, A.V. Joshi, S.G. Davis, A. Laskin, F. Egolfopoulos, C.K. Law, USC Mech Version II. High-temperature combustion reaction model of H2/CO/C1-C4 compounds. http://ignis.usc.edu/USC_Mech_II.htm.
[102] C.W. Gao, J.W. Allen, W.H. Green, R.H. West, Reaction mechanism generator: automatic construction of chemical kinetic mechanisms, Comput. Phys. Commun. 203 (2016) 212–225.
[103] G.R. Magoon, W.H. Green, Design and implementation of a next-generation software interface for on-the-fly quantum and force field calculations in automated reaction mechanism generation, Comput. Chem. Eng. 52 (2013) 35–45.
[104] J.W. Allen, C.F. Goldsmith, W.H. Green, Automatic estimation of pressure-dependent rate coefficients, Phys. Chem. Chem. Phys. 14 (2012) 1131–1155.
[105] A. Cuoci, A. Frassoldati, T. Faravelli, E. Ranzi, OpenSMOKE++: an object-oriented framework for the numerical modeling of reactive systems with detailed kinetic mechanisms, Comput. Phys. Commun. 192 (2015) 237–264.
[106] A. Lifshitz, A. Moran, S. Bidani, Thermal reactions of acetonitrile at high temperatures. Pyrolysis behind reflected shocks, Int. J. Chem. Kinet. 19 (1987) 61–79.
[107] M. Derudi, D. Polino, C. Cavallotti, Toluene and benzyl decomposition mechanisms: elementary reactions and kinetic simulations, Phys. Chem. Chem. Phys. 13 (2011) 21308–21318.
[108] K.O. Johansson, M.F. Campbell, P. Elvati, P.E. Schrader, J. Zádor, N.K. Richards-Henderson, K.R. Wilson, A. Violi, H.A. Michelsen, Photoionization Efficiencies of five polycyclic aromatic hydrocarbons, J. Phys. Chem. A 121 (2017) 4447–4454.
[109] B. Shukla, M. Koshi, A highly efficient growth mechanism of polycyclic aromatic hydrocarbons, Phys. Chem. Chem. Phys. 12 (2010) 2427–2437.
[110] B. Shukla, M. Koshi, A novel route for PAH growth in HACA based mechanisms, Combust. Flame 159 (2012) 3589–3596.
[111] T. Miersch, H. Czech, B. Stengel, G. Abbaszade, J. Orasche, M. Sklorz, T. Streibel, R. Zimmermann, Composition of carbonaceous fine particulate emissions of a flexible fuel DISI engine under high velocity and municipal conditions, Fuel 236 (2019) 1465–1473.
[112] H. Lim, T.M. Ahmed, C. Bergvall, R. Westerholm, Automated clean-up, separation and detection of polycyclic aromatic hydrocarbons in particulate matter extracts using a 2D-LC/2D-GC system: a method translation from two FIDs to two MS detectors, Anal. Bioanal. Chem. 409 (2017) 5619–5629.
[113] Y. Li, L. Zhang, T. Yuan, K. Zhang, J. Yang, B. Yang, F. Qi, C.K. Law, Investigation on fuel-rich premixed flames of monocyclic aromatic hydrocarbons: part I. Intermediate identification and mass spectrometric analysis, Combust. Flame 157 (2010) 143–154.
[114] C. Cavallotti, D. Polino, A. Frassoldati, E. Ranzi, Analysis of some reaction pathways active during cyclopentadiene pyrolysis, J. Phys. Chem. A 116 (2012) 3313–3324.
[115] J.D. Savee, T.M. Selby, O. Welz, C.A. Taatjes, D.L. Osborn, Time- and isomer-resolved measurements of sequential addition of acetylene to the propargyl radical, J. Phys. Chem. Lett. 6 (2015) 4153–4158.
[116] D. Trogolo, A. Maranzana, G. Ghigo, G. Tonachini, First ring formation by radical addition of propargyl to but-1-ene-3-yne in combustion. Theoretical study of the C7H7 radical system, J. Phys. Chem. A 118 (2014) 427–440.
[117] D.S. Parker, R.I. Kaiser, K. Oleg, A. Musahid, Selective formation of indene through the reaction of benzyl radicals with acetylene, ChemPhysChem 16 (2015) 2091–2093.
[118] N.J. Reilly, G. da Silva, C.M. Wilcox, Z. Ge, D.L. Kokkin, T.P. Troy, K. Nauta, S.H. Kable, M.C. McCarthy, T.W. Schmidt, Interconversion of methyltropyl and xylyl radicals: a pathway unavailable to the benzyl–tropyl rearrangement, J. Phys. Chem. A 122 (2018) 1261–1269.
[119] R. Sivaramakrishnan, R.S. Tranter, K. Brezinsky, High pressure pyrolysis of toluene. 2. Modeling benzyl decomposition and formation of soot precursors, J. Phys. Chem. A 110 (2006) 9400–9404.
[120] M. Verdicchio, B. Sirjean, L.S. Tran, P.-A. Glaude, F. Battin-Leclerc, Unimolecular decomposition of tetrahydrofuran: carbene vs. diradical pathways, Proc. Combust. Inst. 35 (2015) 533–541.
[121] K.P. Somers, J.M. Simmie, F. Gillespie, C. Conroy, G. Black, W.K. Metcalfe, F. Battin-Leclerc, P. Dirrenberger, O. Herbinet, P.-A. Glaude, P. Dagaut, C. Togbé, K. Yasunaga, R.X. Fernandes, C. Lee, R. Tripathi, H.J. Curran, A comprehensive experimental and detailed chemical kinetic modelling study of 2,5-dimethylfuran pyrolysis and oxidation, Combust. Flame 160 (2013) 2291–2318.